

Copper in the USA: A Production, Price, and Renewable Energy Lens

A Time Series and Forecasting Analysis

Prepared for Environmental Data Analysis and Visualization

Chhabila Chhatria (MSCC1250017)

Pooja Harish (MSCC1250011)

R. Sai Laxmi (MSCC1250009)

Project Focus

This report independently studies US copper production, global prices, and US-native installed renewable energy capacity, and also looks at the relationships between them. Forecasts have been made for all the variables.

Introduction

Copper is a crucial component of the clean energy transition due to its high durability, conductivity, and recyclability (S&P Global, n.d.). Copper parts and wires are responsible for transmitting the energy required for clean power, EVs, AI data centres and the utilities (electricity) grid. The purpose of our project is to analyse how the United States, a global economic and technological leader who is heavily relying on copper imports among other critical minerals. How has copper production in the US changed over time, and how much of it has been driven by the influence of installed capacity of renewable energy infrastructure and fluctuations in global copper price (traded internationally in USD)?

Using verifiable official data 1990–2024 data from United States Geological Survey (USGS), the International Energy Agency (IEA) and the World Bank, we examine copper across three angles: supply, demand and price. The core of our work is time-series analysis. We have tested each series for the presence of a trend using Mann-Kendall and Theil-Sen's slope, checked stationarity using ADF and KPSS, and performed forecasts till 2030 using ARIMA (since our data is annual, we have avoided SARIMA). Our results have been benchmarked against naive models with validation using back-testing and reasonably wide uncertainty bands given the unpredictability of commercial and economic variables such as the ones we have chosen.

US production shows a slow, price-driven decline (the increase in global copper prices explains more of the decline in production). The decline in production is actually counter-intuitive given the exponential increase

rather than the increase in installed renewable energy capacity registered in the US in the same time period - this means that natively produced US copper is not contributing much at all to native renewable installations. A geographic appendix maps where US copper mines, prospects, processing plants and deposits are situated and where they are concentrated.

****All figures (charts and maps) have been numbered and labelled below the image, not above.**

US Copper Production (1990–2024): Trend Analysis and Forecasting (2025–2030)

1. Trend Analysis

Trend analysis was performed on annual US copper production data spanning 1990–2024, covering a period of nearly 35 years. The objective was to identify and characterise any long-term directional patterns in production.

The pattern of copper production over this period is illustrated in Figure 1 below.

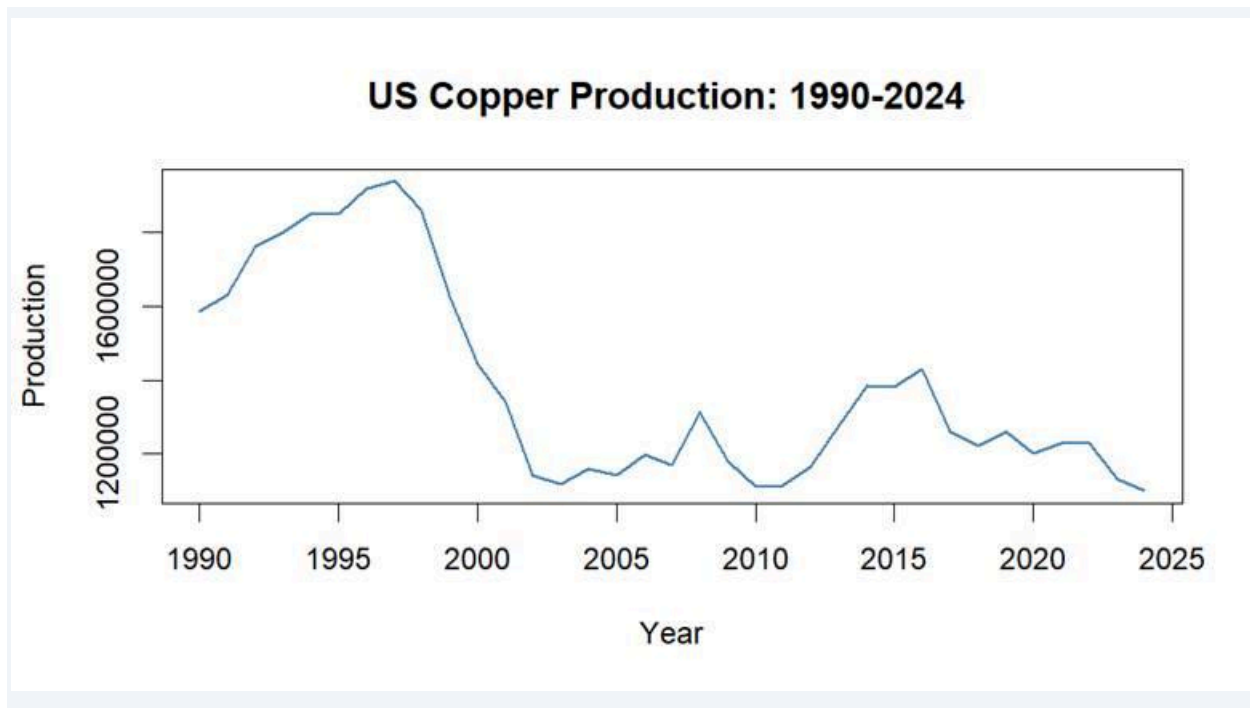


Figure 1. Trend in US copper production between 1990 and 2024.

The graph reveals considerable peaks and troughs in production. To formally test whether a statistically significant trend exists, an Ordinary Least Squares (OLS) regression was performed.

1.1 OLS Regression

The OLS test output is presented below:

Term	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	40,057,506	6,603,090	6.066	7.93e-07 ***
Year	-19,268	3,290	-5.857	1.47e-06 ***

Residuals	Min	1Q	Median	3Q
	-347,314	-165,478	45,708	133,058

R ²	Adjusted R ²	F-statistic (p-value)
0.5097	0.4948	34.3 on 1 & 33 DF (p = 1.47e-06)

The OLS results indicate a statistically significant downward trend in US copper production. However, to assess the reliability of this result, the Durbin-Watson (DW) test was also conducted.

1.2 Durbin-Watson Test for Autocorrelation

DW Statistic	p-value	Conclusion
0.22836	5.981e-15	Strong positive autocorrelation

The DW value close to 0 indicates strong positive autocorrelation in the residuals, which undermines the OLS assumption of independent errors and makes the trend result unreliable. Prewhitening was therefore performed to address this.

1.3 Prewhitening and Trend Significance

Following prewhitening, the trend estimate and significance tests are as follows:

Trend Slope	Kendall Tau	p-value (sig)
-25,237.27	-0.2086	0.0855

Since the p-value (0.0855) exceeds the 0.05 significance threshold, the downward trend is not statistically significant after accounting for autocorrelation. Kendall Tau (-0.21) also indicates only a very weak negative monotonic relationship. The OLS linear trend slope of $-19,268$ is consistent with the prewhitened estimate.

1.4 Stationarity: ACF, PACF, and KPSS Test

The autocorrelation function (ACF) and partial autocorrelation function (PACF) of the copper production time series were plotted to guide model selection (Figure 2).

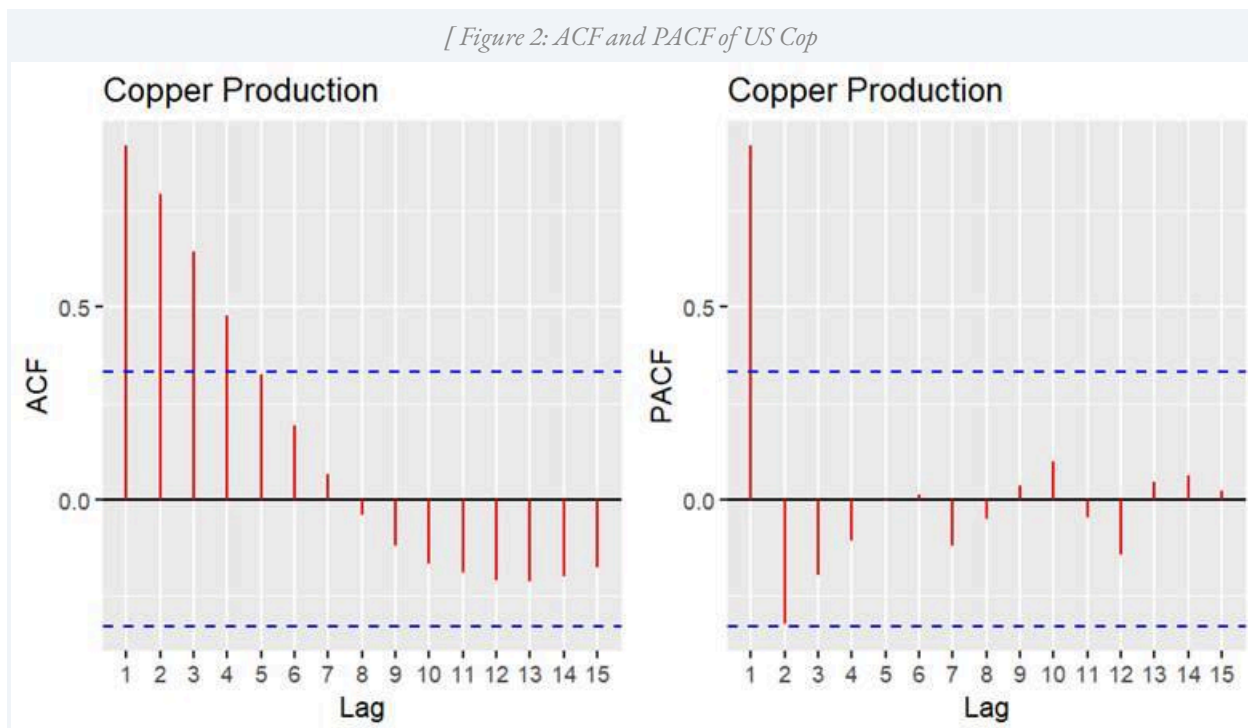


Figure 2. ACF and PACF plots of US copper production. Gradual decay in ACF suggests possible non-stationarity.

The gradual decay in the ACF suggested potential non-stationarity. This was confirmed using the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test:

Test Statistic	10%	5%	2.5%	1%
----------------	-----	----	------	----

0.5798	0.347	0.463	0.574	0.739
--------	-------	-------	-------	-------

The KPSS test statistic (0.5798) exceeds the critical values at the 2.5% and 5% significance levels, indicating evidence against stationarity. Accordingly, a first difference ($d = 1$) was applied in subsequent ARIMA modelling.

2. Forecasting

2.1 ARIMA Model Selection

ARIMA models were built using a first difference ($d = 1$). Candidate models were compared using Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC):

Model	df	AIC	df	BIC
ARIMA(1,1,0)	2	873.57	2	876.62
ARIMA(1,1,1)	3	875.19	3	879.76
ARIMA(1,1,2)	4	876.28	4	882.39
ARIMA(1,1,3)	5	877.48	5	885.11
ARIMA(1,1,4)	6	877.55	6	886.71

ARIMA(1,1,0) was selected as the best-fitting model, having the lowest AIC (873.57) and BIC (876.62) values among the candidates.

2.2 Model Diagnostics

The residuals of the selected ARIMA(1,1,0) model were examined to verify adequacy (Figure 3).

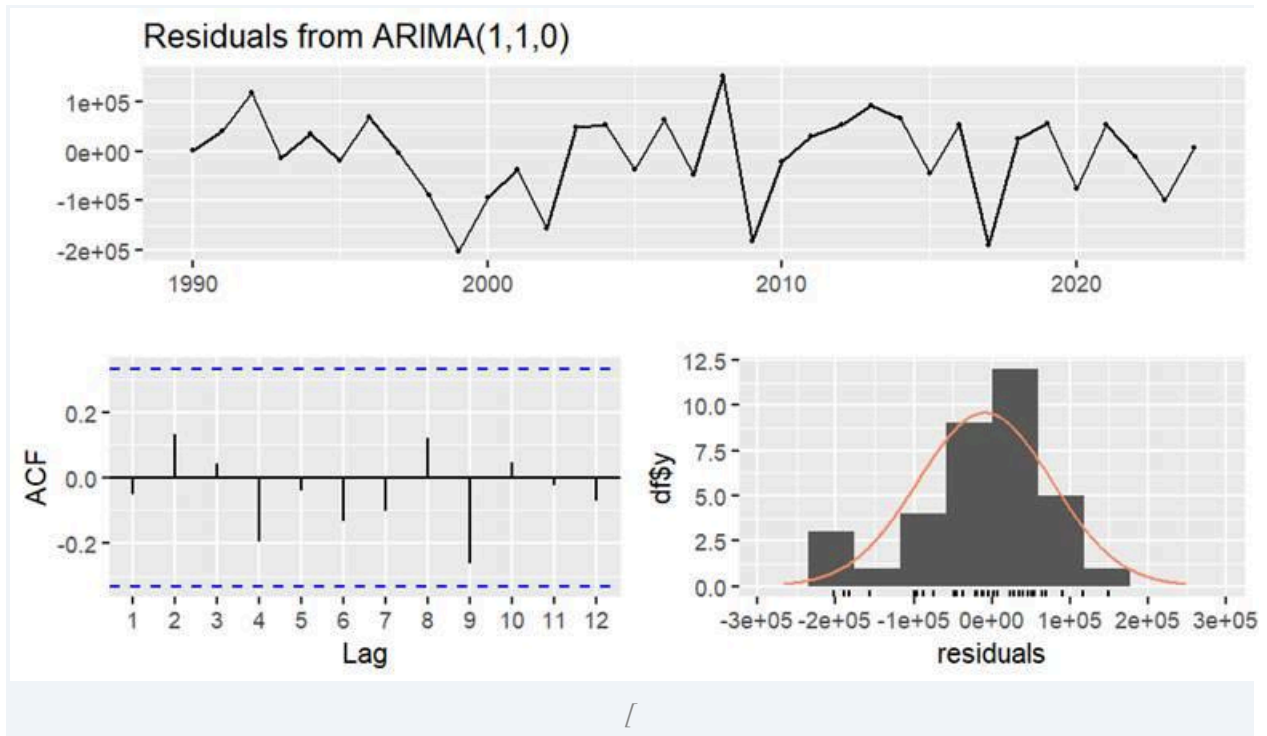


Figure 3. Residual diagnostics for the ARIMA(1,1,0) model. No significant spikes in the ACF of residuals, indicating the model adequately captures the underlying pattern.

2.3 Model Validation (Training/Testing)

The model was validated against a held-out test set (Figure 4):

Training Period	Test Period
1990–2020	2021–2024

Figure 4. Model forecast vs. observed values during the training (1990–2020) and test (2021–2024) periods. Actual values remain within the confidence interval; model outperforms the naive forecast.

The model's predictions fell within the confidence interval during the test period and outperformed the naive forecast, confirming suitability for future projection.

2.4 Five-Year Forecast (2025–2030)

The validated ARIMA(1,1,0) model was used to forecast US copper production from 2025 to 2030 (Figure 5 and Table below).

Figure 5. Point forecasts and 80%/95% confidence intervals for US copper production, 2025–2030.

Year	Point Forecast	Lo 80%	Hi 80%	Lo 95%	Hi 95%	
2025	1,088,783	976,401	1,201,166	916,909	1,260,658	
2026	1,084,590	893,620	1,275,559	792,527	1,376,652	
2027	1,083,022	827,274	1,338,770	691,889	1,474,154	
2028	1,082,435	771,988	1,392,883	607,647	1,557,224	
2029	1,082,216	724,270	1,440,163	534,784	1,629,648	
2030	1,082,134	681,925	1,482,343	470,067	1,694,201	

Forecasts indicate that copper production will remain relatively stable, showing only a very slight decline across 2025–2030.

3. Interpretation and Inference

The trend analysis reveals that US copper production does not exhibit a statistically significant monotonic trend and instead fluctuates around a relatively stable level. The five-year forecast is consistent with this pattern, projecting broadly stable output through 2030.

From a real-world perspective, these findings are characteristic of a mature resource-based industry. The observed weak downward tendency may be attributed to:

- Depletion of high-grade ore reserves
- Rising extraction costs
- Environmental regulation
- Global competition from major copper-producing nations
- Cyclical commodity market dynamics (Díaz, 2015; Wang et al., 2015; Northey et al., 2014)

Overall, US copper production appears to be stabilising at a mature stage with mild long-term downward pressure. The ARIMA(1,1,0) model provides reliable forecasts and captures the underlying time-series structure effectively, making it a suitable tool for projecting near-term production trends.

4. Correlation with Copper Price and Renewable Energy Capacity

To explore potential drivers of the observed production trend, global copper price and installed renewable energy capacity in the US were examined. A Pearson correlation matrix was generated across the three variables.

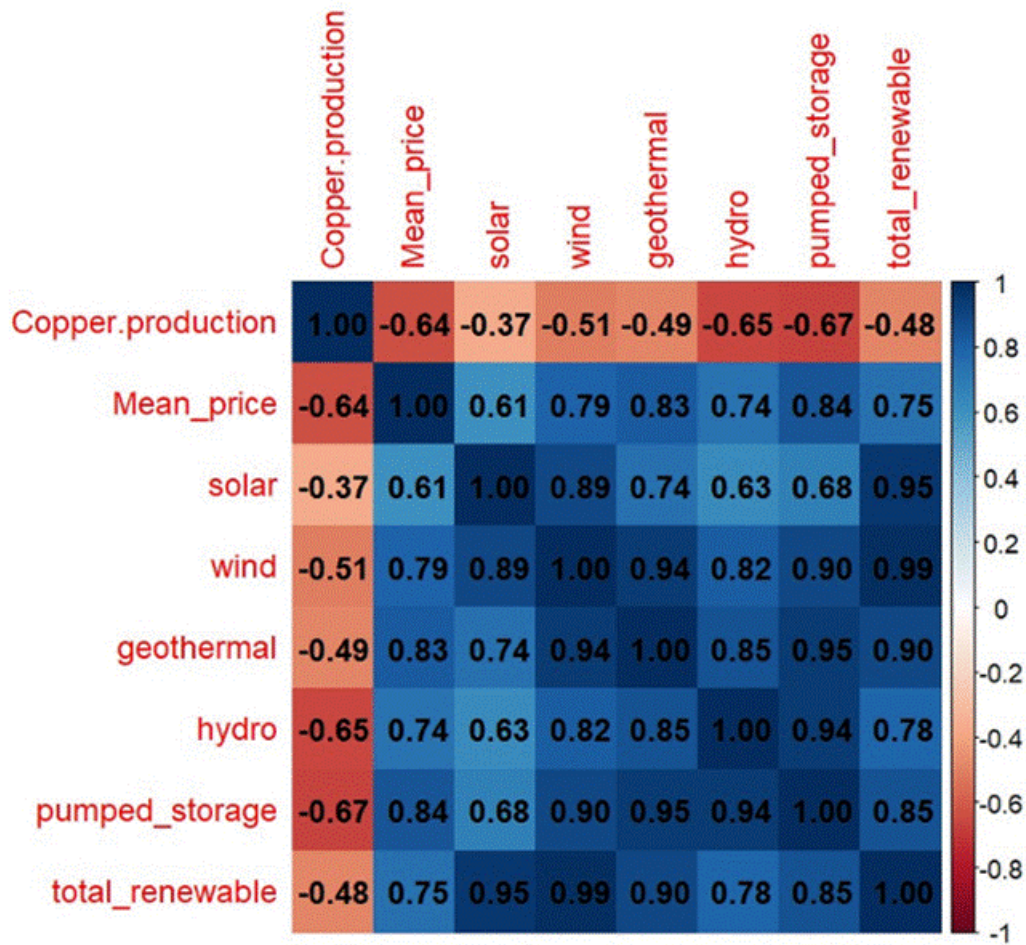


Figure 6. Pearson correlation matrix showing the relationships between US copper production, mean global copper price, and installed renewable energy capacity.

Key correlation findings:

- US copper production vs. renewable energy capacity: negative correlation
- US copper production vs. mean copper price: positive correlation

5. Regression Analysis

A multiple regression analysis was conducted with copper production as the dependent variable (Y), and mean copper price (X_1) and installed renewable energy capacity (X_2) as independent variables.

5.1 Effect of Mean Copper Price

Copper production is strongly and significantly influenced by mean price ($\beta = -65.50$, $p = 3.59e-05$). This indicates that higher prices are associated with lower production levels over the study period. Price alone explains a moderate share of production variance ($R^2 = 0.41$).

5.2 Effect of Renewable Energy Capacity

Renewable energy capacity shows a significant univariate relationship with production ($\beta = -0.407$, $p = 0.0035$, $R^2 = 0.23$). However, when included alongside price in the combined model, its effect disappears entirely ($\beta = -0.0026$, $p = 0.988$), indicating that this relationship is not independent of price effects.

5.3 Combined Model

Variable	β (Estimate)	p-value	R^2 (model)	Significance
Mean Copper Price (X_1)	-65.27	0.004	0.41	Significant
Renewable Energy (X_2)	-0.0026	0.988	0.41	Not significant

The combined model confirms that mean price is the primary explanatory factor for copper production. Renewable energy does not improve the model once price is included. This suggests the system is principally driven by economic forces rather than structural changes linked to the energy transition. While renewable energy shows some relationship in isolation, it lacks independent explanatory power when modelled alongside price.

References for US Copper Production

- Díaz, M. (2015). The rise and fall of an industry: Entry in U.S. copper mining, 1835–1986. *Resource and Energy Economics*, 42, 141–169. <https://doi.org/10.1016/j.reseneeco.2015.08.001>
- Wang, M., Chen, W., & Li, X. (2015). Substance flow analysis of copper in production stage in the U.S. from 1974 to 2012. *Resources, Conservation and Recycling*, 105, 36–48. <https://doi.org/10.1016/j.resconrec.2015.10.012>
- Northey, S., Mohr, S., Mudd, G. M., Weng, Z., & Giurco, D. (2014). Modelling future copper ore grade decline based on a detailed assessment of copper resources and mining. *Resources, Conservation and Recycling*, 83, 190–201. <https://doi.org/10.1016/j.resconrec.2013.10.005>

Domestic Copper Demand from US Renewable Installed Capacity

Background of copper demand from US renewable capacity growth

As the United States expands renewable energy capacity, demand for copper is expected to increase.

This section focuses on the renewable-energy demand side of copper use. Instead of analysing copper production or price directly, this study examines how US renewable energy capacity has changed over time and estimates how much copper may be needed for future solar and wind expansion.

Installed capacity is used as the main demand indicator because copper is required when infrastructure is built. Therefore, new additions in solar and wind capacity are directly linked to potential copper demand.

1. Research Objectives

1. Analyse the growth of US renewable energy capacity from 1990 to 2024.
2. Identify which renewable sources are driving this growth.
3. Statistically test whether renewable capacity shows significant trends.
4. Forecast total renewable, solar, and wind capacity up to 2030.
5. Estimate future copper demand from new solar and wind capacity additions.

2. Data Description

The dataset contains annual US renewable energy capacity by source from 1990 to 2024. The source is the US Energy Information Administration. The original unit is megawatts (MW), and values were converted to gigawatts (GW) for visualization and interpretation.

Item	Description
Data file	Installedcapacity_Annual - Co.csv
Years	1990-2024
Observations	35 annual observations
Original unit	Megawatts (MW)
Plot/report unit	Gigawatts (GW), where 1 GW = 1,000 MW
Variables	Year, solar, wind, geothermal, hydro, pumped_storage, total_renewable

3. Methodology

The analysis followed a step-by-step workflow:

6. Load and inspect the renewable capacity data.
7. Validate data completeness, year continuity, duplicate years, and total renewable calculations.
8. Convert MW to GW and reshape the dataset from wide to long format for visualization.
9. Create exploratory visualizations to understand total growth, source composition, and solar-wind additions.
10. Use Mann-Kendall and Theil-Sen's slope to test and quantify long-term trends.
11. Convert total, solar, and wind capacity into time series objects.
12. Test stationarity using ADF, KPSS, and ndiffs().
13. Fit ARIMA models using train-test validation.
14. Compare solar ARIMA with naive, drift, ETS, and damped ETS models because solar had higher uncertainty.
15. Refit final models on the full 1990-2024 data and forecast to 2030.
16. Convert forecasted new solar and wind capacity into copper demand using copper intensity assumptions.

Why ACF/PACF are not shown as final figures

ACF and PACF are useful for manually selecting ARIMA parameters. In this project, `auto.arima()` was used to compare candidate ARIMA models automatically. Therefore, stationarity testing, train-test validation, and forecast accuracy were emphasized instead of manual ACF/PACF selection.

Why Random Forest was not used

Random Forest was covered in class for AQI forecasting, but this dataset has only 35 annual observations and few predictor variables. Random Forest usually needs more observations and richer features, so ARIMA and benchmark time-series models were more defensible here.

4. Data Cleaning and Validation

The data contained 35 rows and 7 columns. The year sequence was continuous from 1990 to 2024, and no duplicate years were found. The total renewable capacity column was checked against the sum of solar, wind, geothermal, hydro, and pumped storage.

Validation Check	Result	Interpretation
Duplicate years	TRUE for no duplicates	Each year appears only once.
Continuous years	TRUE for 1990-2024	No missing year in the sequence.
Total renewable check	Maximum difference: 2.328306e-10	Difference is essentially zero; tiny value is computer rounding.
Conclusion	Data is clean	Suitable for analysis.

5. Exploratory Data Analysis

Exploratory visualization was used to understand the historical pattern before formal trend testing and forecasting.

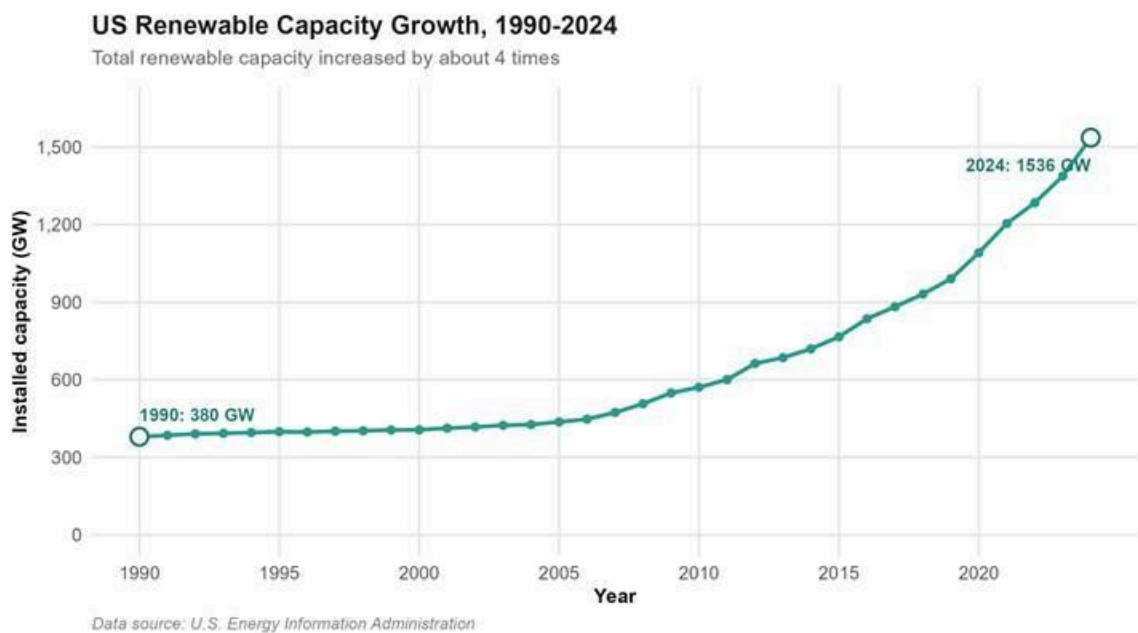


Figure 1. US renewable capacity growth, 1990-2024.

Total US renewable capacity increased from about 380 GW in 1990 to about 1,536 GW in 2024. This is roughly four times growth over the study period.

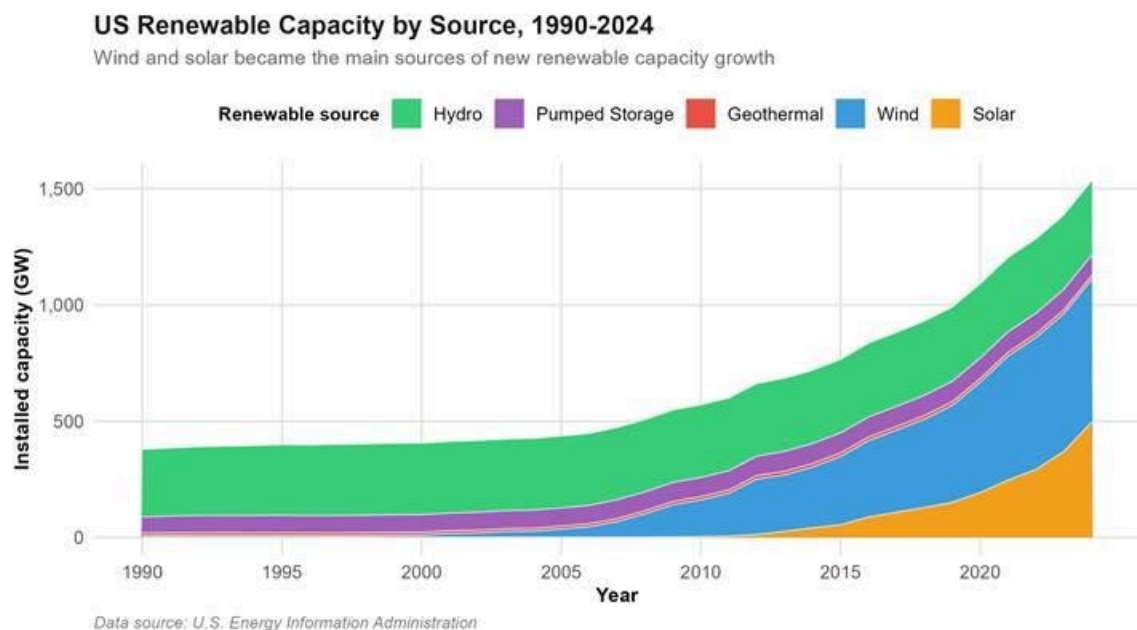


Figure 2. Renewable capacity by source, 1990-2024.

Hydro dominated renewable capacity in the early years, but hydro capacity remained relatively stable while wind and solar increased rapidly.

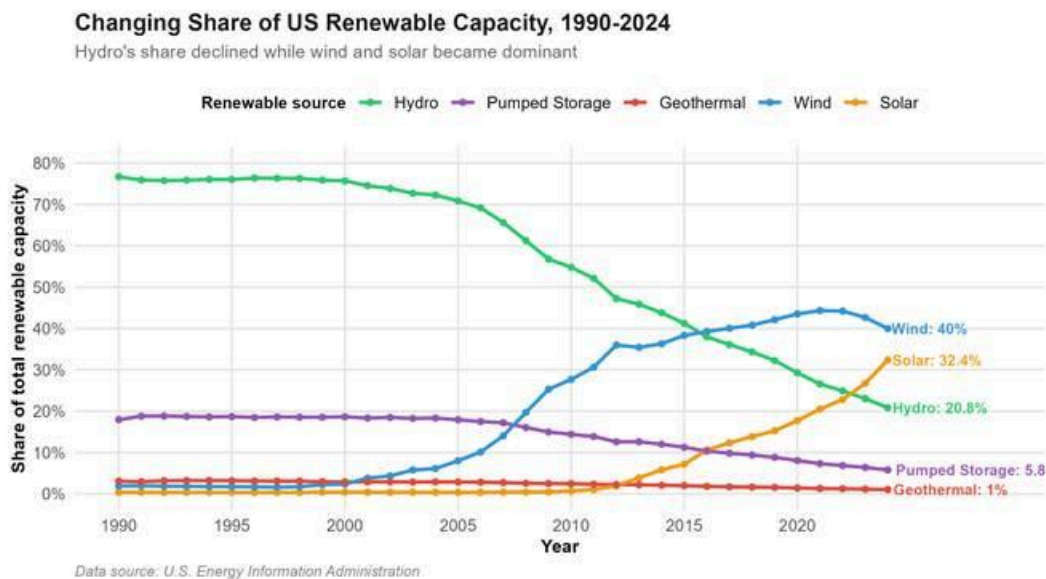


Figure 3. Changing share of renewable capacity by source, 1990-2024.

The renewable mix changed strongly. Hydro's share declined, while wind and solar became increasingly important in the renewable capacity mix.

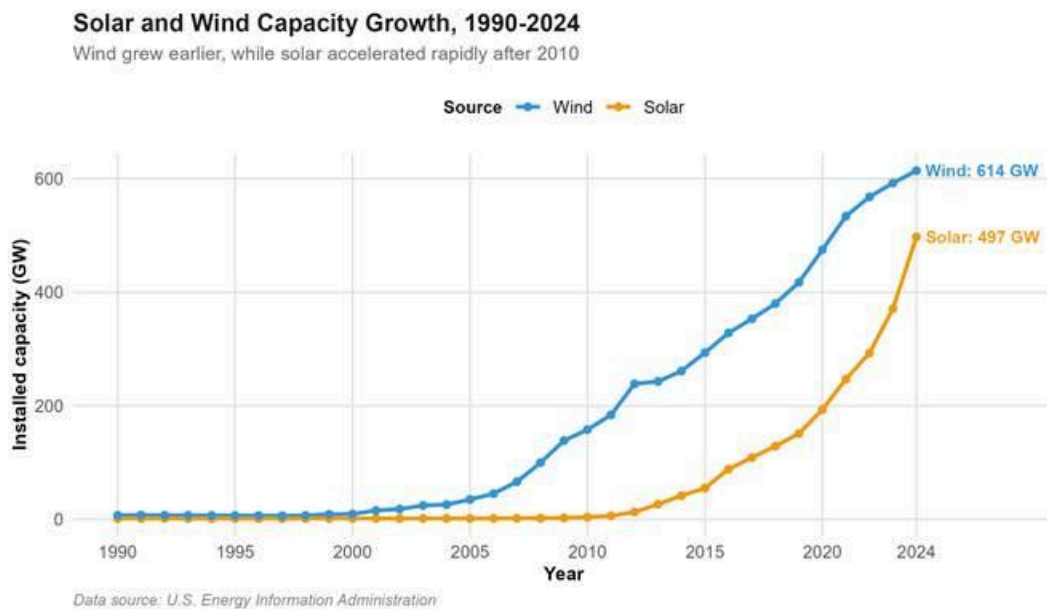


Figure 4. Solar and wind capacity growth, 1990-2024.

Solar and wind were examined separately because they are major copper-demand drivers. Wind grew earlier, while solar accelerated strongly after 2010.

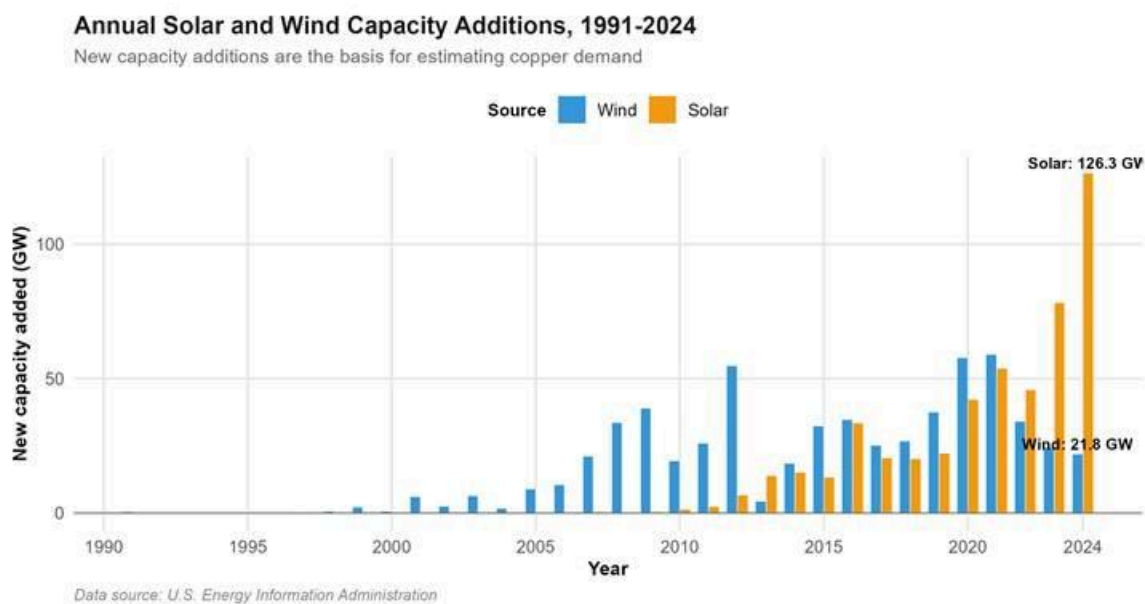


Figure 5. Annual solar and wind capacity additions, 1991-2024.

Annual additions are significant because copper is required when new capacity is built. The plot shows sharp increases in recent solar additions.

6. Trend Analysis

The Mann-Kendall test was used to test whether each renewable source had a statistically significant trend. Theil-Sen's slope was used to estimate the growth rate in GW per year.

Source	Tau	P-value	Theil-Sen slope (GW/year)	Significant
Total Renewable	0.997	<2e-16	25.19	Yes
Solar	0.855	7.11e-13	3.81	Yes
Wind	0.892	5.2e-14	18.42	Yes
Geothermal	0.789	3.26e-11	0.14	Yes
Hydro	0.970	2.84e-16	0.66	Yes
Pumped Storage	0.969	4.78e-16	0.56	Yes

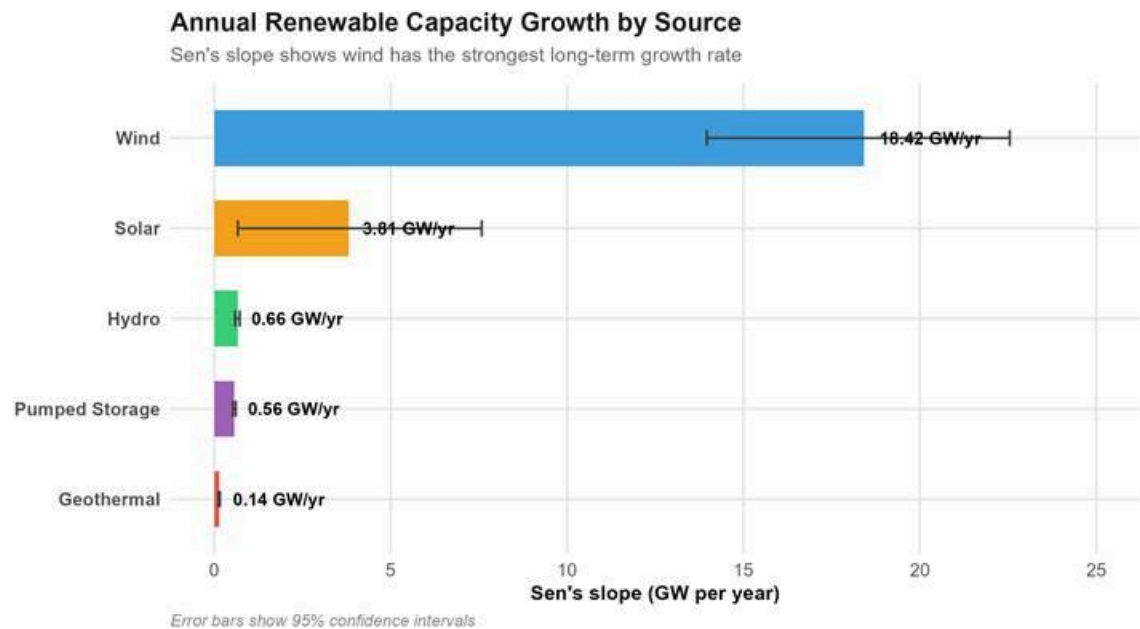


Figure 6. Annual renewable capacity growth by source using Theil-Sen's slope.

All sources had significant upward trends. Wind had the strongest long-term annual growth rate. Solar's long-term slope is lower because it was very small for many early years, but it accelerated strongly later.

7. Stationarity Testing

Before ARIMA forecasting, stationarity was tested using ADF, KPSS, and ndiffs(). The original series were non-stationary because capacity keeps increasing over time.

Series	ADF level p	KPSS level p	ADF diff1 p	KPSS diff1 p	ndiffs
Total Renewable	0.9900	0.0100	0.990	0.010	2
Solar	0.9900	0.0164	0.990	0.014	2
Wind	0.9022	0.0100	0.332	0.010	2

The `ndiffs()` result suggested that second differencing was needed, which matched the selected ARIMA models.

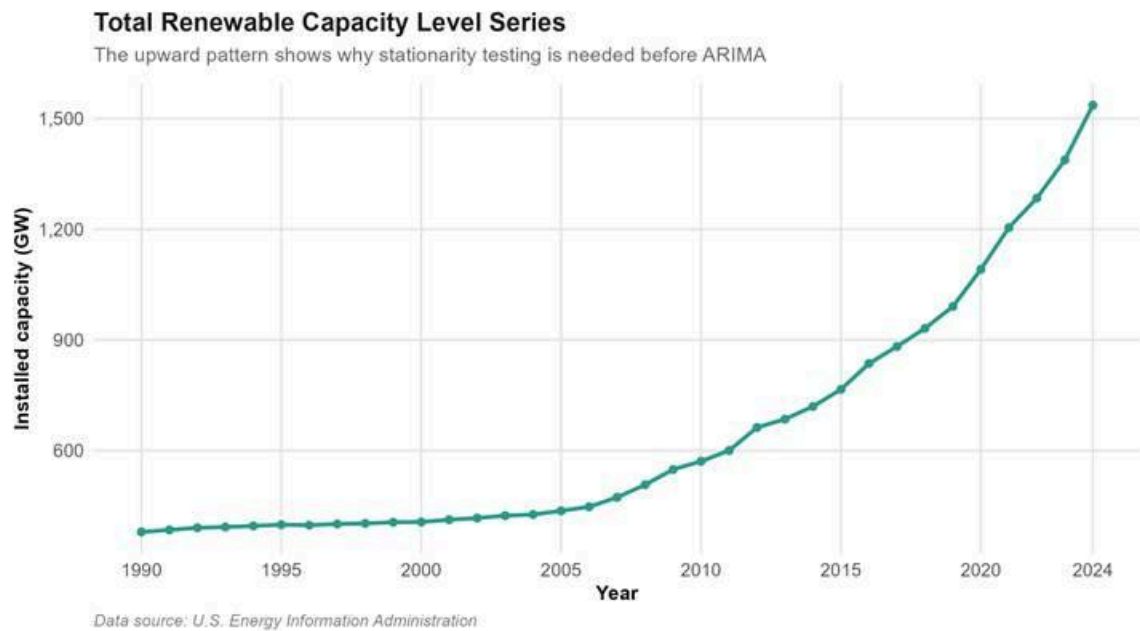


Figure 7. Total renewable capacity level series.

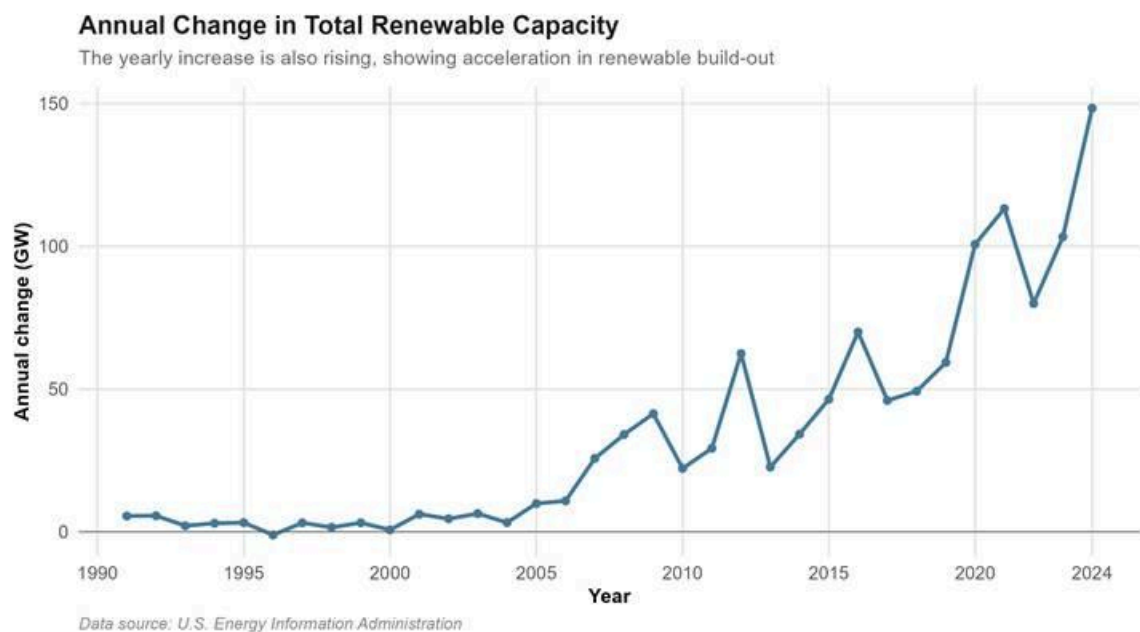


Figure 8. Annual change in total renewable capacity.

8. Forecasting and Model Evaluation

ARIMA models were first trained using data from 1990 to 2018 and tested on 2019 to 2024. This train-test split checks whether the model can predict years it has not seen.

Series	Training ARIMA Model	Test MAPE	Decision
Total Renewable	ARIMA(0,2,1)	10.59%	Acceptable
Solar	ARIMA(1,2,0)	25.52%	Needs alternative check
Wind	ARIMA(0,2,1)	10.45%	Acceptable

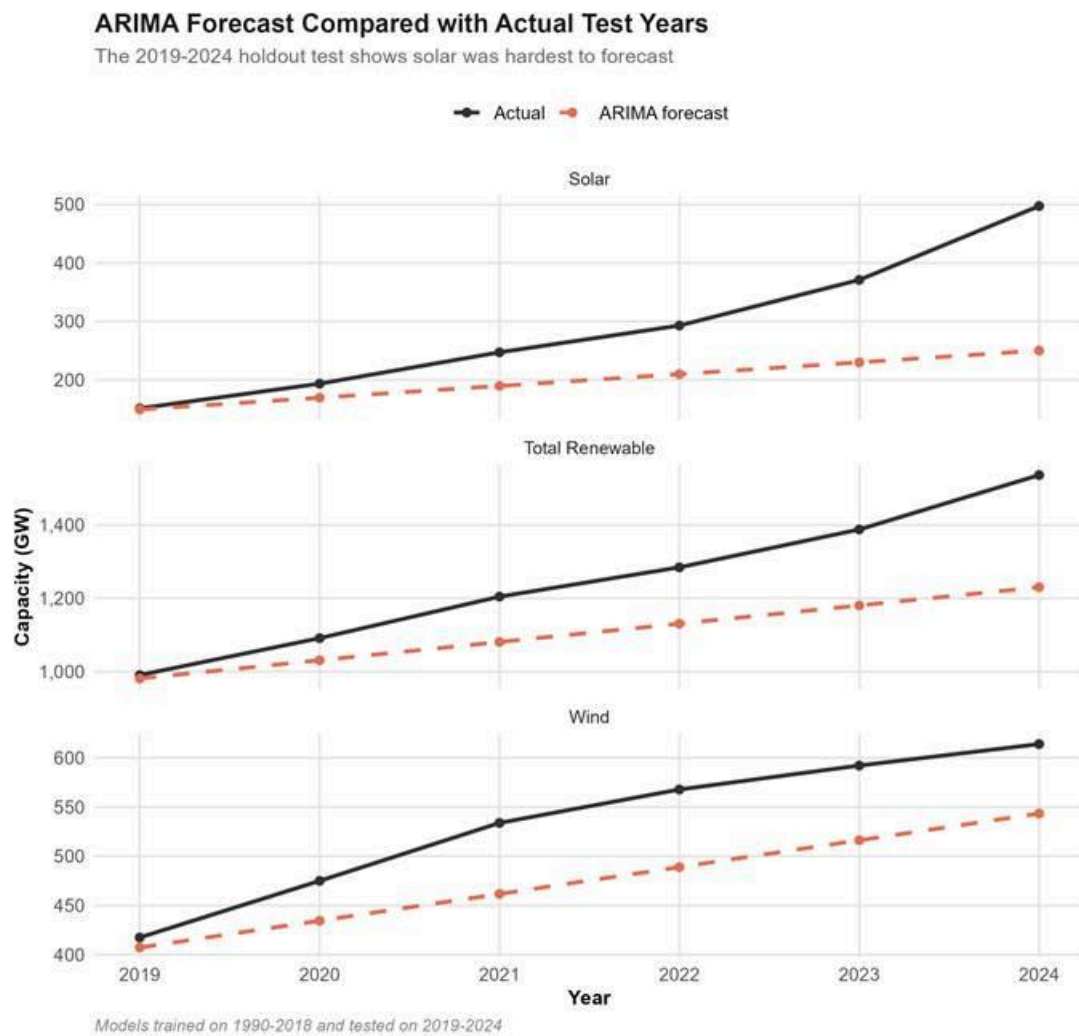


Figure 9. ARIMA forecast compared with actual test years, 2019-2024.

ARIMA captured the general upward direction but underestimated recent growth, especially solar. Solar was difficult because it accelerated sharply after 2019.

9. Solar Alternative Model Testing

Because solar ARIMA had high error, it was compared with naive, drift, ETS, and damped ETS models.

Model	RMSE	MAE	MAPE
ARIMA	123680.6	92473.0	25.52%
ETS	123928.3	92713.3	25.60%
Damped ETS	128435.7	96376.2	26.67%
Drift	182472.6	147188.4	43.32%
Naive	199834.2	163126.0	48.51%

ARIMA remained the best model among the tested options. ETS was almost the same as ARIMA, while drift and naive performed worse.

Train Period	Test Period	Model	Test Years	MAPE
1990-2016	2017-2024	ARIMA(0,2,0)	8	14.83%
1990-2018	2019-2024	ARIMA(1,2,0)	6	25.52%
1990-2020	2021-2024	ARIMA(0,2,0)	4	12.67%
1990-2021	2022-2024	ARIMA(0,2,0)	3	8.36%

Changing the train-test period showed that solar accuracy improved when recent acceleration was included in training. This means the original solar weakness was mainly caused by rapid structural acceleration, not a coding mistake.

Final Solar Model	2030 Forecast (GW)	Lower 95% (GW)	Upper 95% (GW)
ARIMA	1342.4	1061.0	1623.7
ETS	1255.2	-1306.1	3816.6
Damped ETS	1203.9	-1327.1	3734.9

ETS and damped ETS produced negative lower confidence bounds, which are not physically meaningful for installed solar capacity. Therefore, ARIMA was retained as the final solar model.

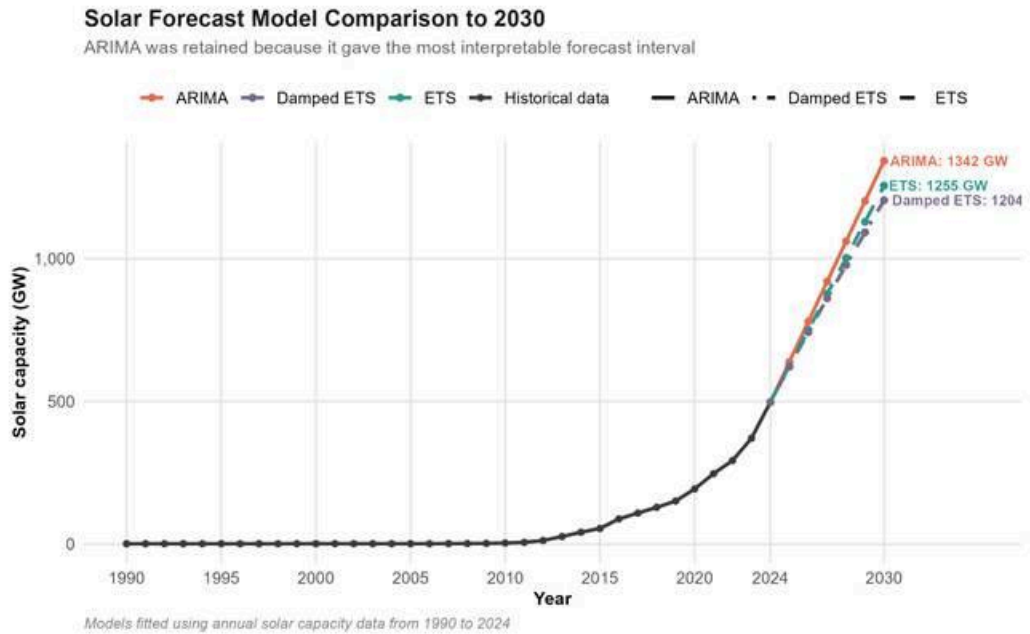


Figure 12. Solar forecast model comparison to 2030. This is a backup/sensitivity figure.

10. Final Forecast to 2030

The final ARIMA models were refit using the full 1990-2024 dataset and used to forecast capacity to 2030.

Series	2024 Capacity (GW)	2030 Forecast (GW)	Lower 95% (GW)	Upper 95% (GW)
Total Renewable	1536.0	2426.3	2090.0	2762.6
Solar	497.2	1342.4	1061.0	1623.7
Wind	614.0	784.7	644.6	924.8

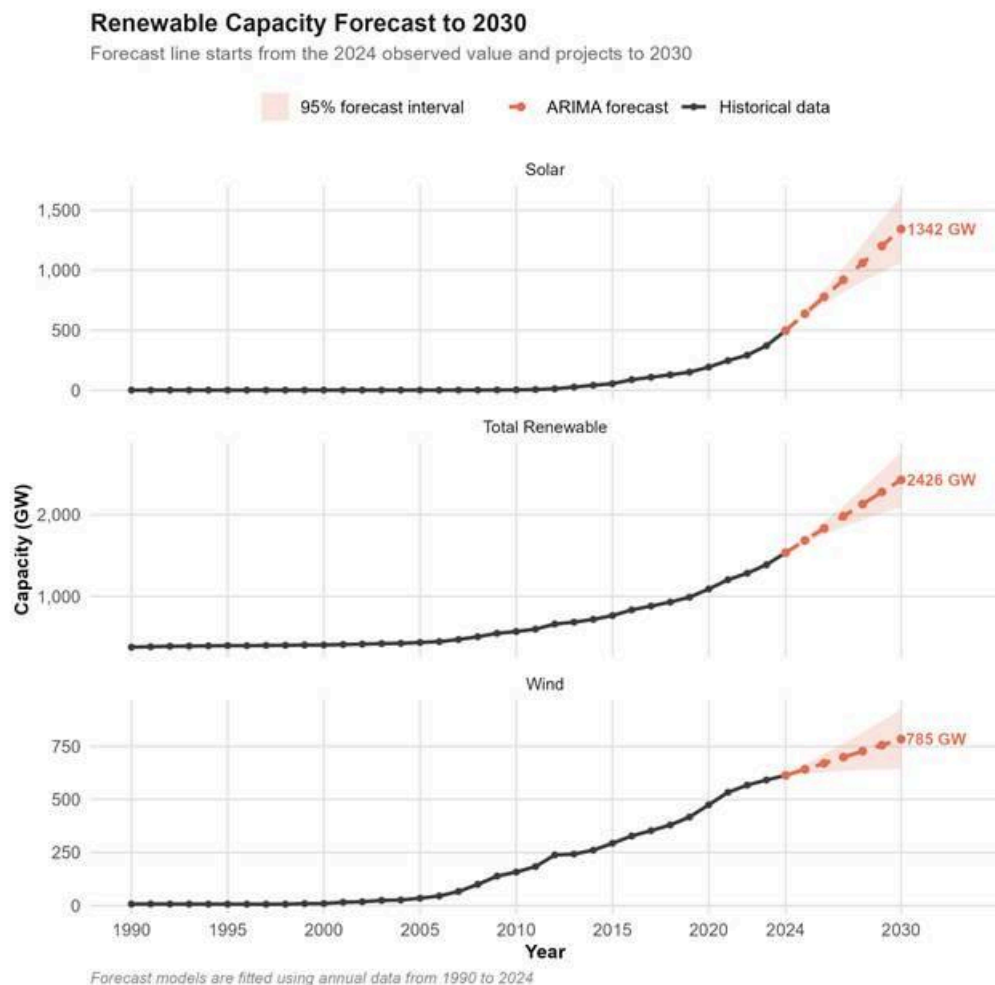


Figure 10. Renewable capacity forecast to 2030.

The forecast suggests continued renewable growth, with solar showing the largest future increase.

11. Copper Demand Estimation

Copper demand was estimated from new solar and wind capacity additions, not from total existing capacity.

The formula was:

Copper demand formula

New capacity added (MW) x copper intensity (tonnes/MW) = copper demand (tonnes)

Technology	Copper intensity assumption
Solar	5 tonnes/MW
Wind	3 tonnes/MW

Year	Solar added (GW)	Wind added (GW)	Total copper (tonnes)
2025	140.9	28.5	789648
2026	140.9	28.5	789648
2027	140.9	28.5	789648
2028	140.9	28.5	789648
2029	140.9	28.5	789648
2030	140.9	28.5	789648

Total estimated copper demand from new solar and wind capacity from 2025 to 2030 is 4,737,888 tonnes, or approximately 4.74 million tonnes.

The annual copper values are similar because the ARIMA forecast projects nearly steady annual additions from 2025 to 2030. This is best interpreted as an average annual model-based requirement, not an exact real-world yearly prediction.

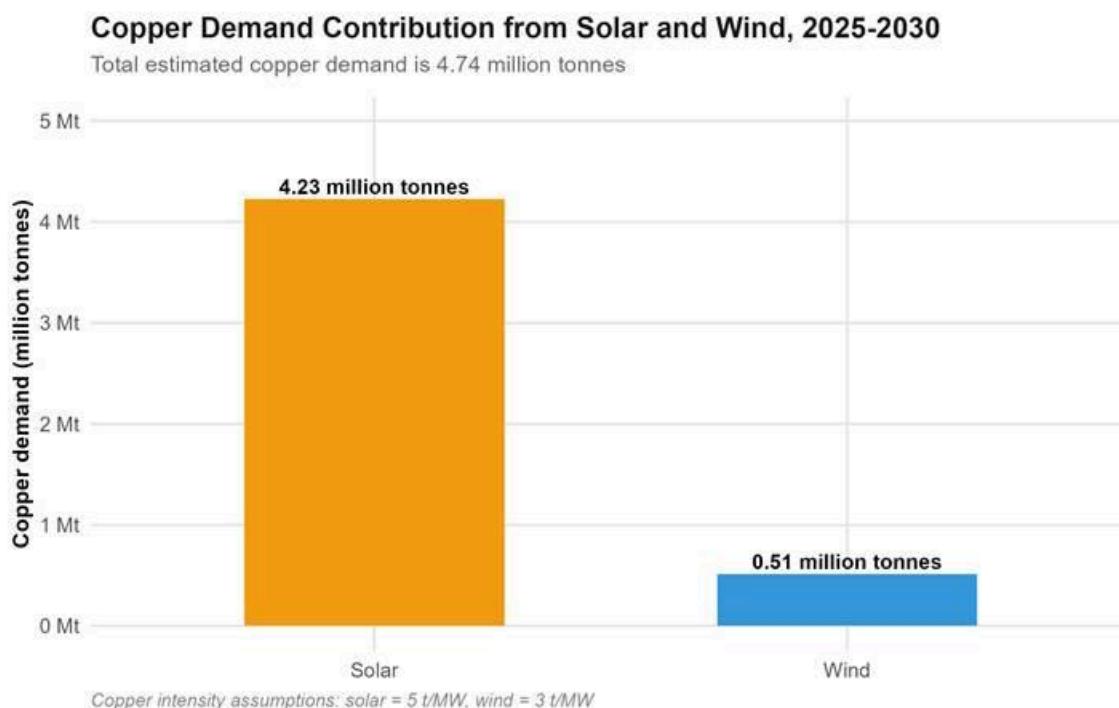


Figure 11. Copper demand contribution from solar and wind, 2025-2030.

12. Discussion

The results show that renewable energy expansion can create a substantial pull on copper demand. The growth is not evenly distributed across all renewable sources. Hydro was historically dominant but remained relatively stable, while wind and solar became the main drivers of new capacity growth.

Solar is especially important for future copper demand because the forecast suggests strong capacity additions through 2030. However, solar forecasting also has higher uncertainty because recent solar growth accelerated rapidly.

The model comparison showed that ARIMA remained the most defensible model for this dataset. ETS and damped ETS did not improve accuracy and produced physically unrealistic negative lower forecast intervals for solar. Naive and drift models performed worse than ARIMA.

13. Limitations

- ARIMA uses past patterns and does not directly include future policy changes, subsidies, investment shifts, technology breakthroughs, or supply-chain constraints.
- Solar forecast uncertainty is higher because solar growth accelerated sharply after 2019.
- Copper demand was estimated using fixed copper intensity assumptions.
- Actual copper use can vary depending on technology design, turbine type, solar panel type, cable length, and grid connection needs.
- The copper estimate includes only solar and wind capacity additions. It does not include copper demand from grid expansion, battery storage, electric vehicles, data centres, buildings, or other electrification infrastructure.
- The dataset contains only 35 annual observations, which limits the complexity of forecasting models that can be used.

14. Inference

US renewable energy capacity increased strongly from 1990 to 2024, growing from about 380 GW to 1,536 GW. The renewable mix also changed over time. Hydro was dominant in the early years, but wind and solar have become the main sources of growth in recent years.

Trend analysis confirmed statistically significant upward trends across all renewable sources. Theil-Sen's slope showed that wind had the strongest long-term growth rate, while solar showed strong recent acceleration.

ARIMA forecasting suggests that by 2030, total renewable capacity may reach about 2,426 GW, solar may reach about 1,342 GW, and wind may reach about 785 GW.

Using copper intensity assumptions, new solar and wind capacity from 2025 to 2030 may require about 4.74 million tonnes of copper. Most of this estimated copper demand comes from solar expansion. Overall, the analysis supports the idea that the clean-energy transition can increase demand for critical minerals such as copper.

15. Figure Checklist

Figure	Title	Placement
1	US Renewable Capacity Growth, 1990-2024	Exploratory Data Analysis
2	Renewable Capacity by Source, 1990-2024	Exploratory Data Analysis
3	Changing Share of Renewable Capacity, 1990-2024	Exploratory Data Analysis
4	Solar and Wind Capacity Growth, 1990-2024	Exploratory Data Analysis
5	Annual Solar and Wind Capacity Additions, 1991-2024	Exploratory Data Analysis
6	Annual Renewable Capacity Growth by Source	Trend Analysis
7	Total Renewable Capacity Level Series	Stationarity Testing
8	Annual Change in Total Renewable Capacity	Stationarity Testing
9	ARIMA Forecast Compared with Actual Test Years	Forecast Evaluation
10	Renewable Capacity Forecast to 2030	Final Forecast
11	Copper Demand Contribution from Solar and Wind, 2025-2030	Copper Demand Estimation
12	Solar Forecast Model Comparison to 2030	Backup/Sensitivity

16. References for copper demand from US renewable capacity growth

- US Energy Information Administration. Annual renewable electricity capacity data.
- Seck, G. S., Hache, E., Bonnet, C., Simoen, M., & Carcanague, S. (2020). Copper at the crossroads: Assessment of the interactions between low-carbon energy transition and supply limitations. *Resources, Conservation and Recycling*, 163, 105072.
- Prakash, C., Dhyani, B., Chauhan, A., & Sayal, A. (2025). ARIMA based forecasting of solar and hydro energy consumption with implications for grid stability and renewable policy. *Discover Sustainability*, 6, 663.
- Ozturk, S., & Ozturk, F. (2018). Forecasting energy consumption of Turkey by ARIMA model. *Journal of Asian Scientific Research*, 8(2), 52-60.
- S&P Global. (n.d.). *Copper in the age of AI*.
<https://www.spglobal.com/en/research-insights/special-reports/copper-in-the-age-of-ai>

Global Copper Prices and USA Copper Sites

Geographic Distribution

Pooja Harish • MTH-602 Environmental Data Analysis • Annual series 1990–2024, forecasts to 2030

Background of Copper Price and Spatial Visualisation

Copper's global price captures and expresses the world's demands and anticipation regarding the value of copper in the ongoing energy transition from fossil-fuel to renewable. Using the World Bank "Pink Sheet" benchmark (annual, 1990–2024), this analysis examines whether copper's sharp post-2020 rise is a real, lasting shift or part of a short-lived commodity cycle which dies out over time. Nominal price has been deflated with two yardsticks — US consumer prices (CPI) and global manufactured-goods prices (MUV) — then the trend has been tested for strength, non-stationarity has been verified, and forecasts have been drawn to 2030 with ARIMA, keeping the wide uncertainty bands transparent.

Time-Series Analysis

1. Nominal vs. real (inflation-adjusted) prices

The **nominal** price includes both inflation (value of money lost over years) and the actual commodity price of copper - since inflation can obscure the underlying trend of the commodity price, I have chosen to deflate the prices using 2 different variables. **Deflating** divides the price by a price index to remove the effect of inflation. The two indices used are the CPI and the MUV - CPI being a US Bureau of Labour Statistics measure, and the MUV being a World Bank published index. The CPI tracks how expensive end consumer goods in the US have grown over the years - it includes a wide range of goods including groceries, automobiles, electronics, transportation, medical, care, etc. (U.S. Bureau of Labor Statistics, 2020). CPI has been used since copper is embedded in a wide range of consumer goods such as electronics and digital devices, the cost of electricity & transportation, which themselves partially rely on components made of copper etc. (Cuddington & Jerrett, 2008). The **CPI** is calculated as follows, directly quoted from the U.S. Bureau of Labor Statistics (2020) :

The price relative (using a geometric mean formula) is given by:

Geometric price relative:

$$R^G_{i,t-1,t} = [\prod_{j=1}^m (p_{j,t} / p_{j,t-1})]^{1/m}$$

Laspeyres price relative:

$$R^L_{i,a,t-1,t} = [\sum_{j=1}^m w_{ja0}(p_{jat} / p_{ja0})] / [\sum_{j=1}^m w_{ja0}(p_{ja,t-1} / p_{ja0})]$$

$R^G_{i,a,t-1,t}$ = geometric price relative for item-area combination (i,a) from period t–1 to period t

$R^L_{i,a,t-1,t}$ = Laspeyres price relative for item-area combination (i,a) from period t–1 to period t

p_{jat} = price of item j in area a observed in period t

$p_{ja,t-1}$ = price of the same item j in area a observed in period t–1

p_{ja0} = estimated price of item j in the base period

w_{ja0} = base-period weight of item j

m = number of usable price quotes for the item-area combination (i,a)

The product and sums in the formulas presented above are taken over all price quotes which are usable for estimation in the item-area combination (i,a). It is important that the price of each quote be collected (or estimated) in both periods in order to measure price change (U.S. Bureau of Labor Statistics, 2020).

The **MUV** is a composite index which tracks the changes in export value of 1 unit of a commodity manufactured in any of the G5 industrialised countries (insert reference) (World Bank, n.d.). This deflator has been used since copper is an industrial output which acts as a raw material for a wide range of goods - both infrastructural (such as electricity grids, cables and wires, renewable energy etc.) and consumable (batteries, electric vehicles, appliances, etc.) (Grilli & Yang, 1988). The formula for MUV is (World Bank, 1997) (Moran & Park, 1986):

The MUV is a weighted average of individual country indices, typically calculated using a Paasche or Fisher index framework. Conceptually, the formula is calculated as follows:

$$MUV = \sum_{i=1}^n (W_i \times UV_i)$$

Where:

- W_i = export weight of country i
- UV_i = unit value index of manufactured exports for country i
- n = number of major exporting countries, historically G-5: France, Germany, Japan, UK, USA

MUV is published on an annual basis but CPI is measured on a monthly basis, hence to convert to annual, the **average of the monthly values** was taken for each year.

The formulas for performing the nominal price deflations were:

CPI-adjusted copper price = Nominal copper price * CPI value in 2010 / CPI value in that year

MUV-adjusted copper price = Nominal copper price * MUV value in 2010 / MUV value in that year

Since the value of the dollar has been changing throughout this time, the assumption to make all the price trends comparable over time was:

cpi_reference_value = CPI value in 2010

muv_reference_value = MUV value in 2010

Since both are expressed in 2010-adjusted dollars, the interpretation is: **How much would the nominal copper price be worth if converted into 2010 price-level terms?** This logic of 2010 is compatible with the World Bank's choice of taking 2010 = 100 as the base year in computing their yearly MUV, as can be seen in their pink sheet data, which means that all the MUV values are expressed as a ratio of the MUV in 2010.

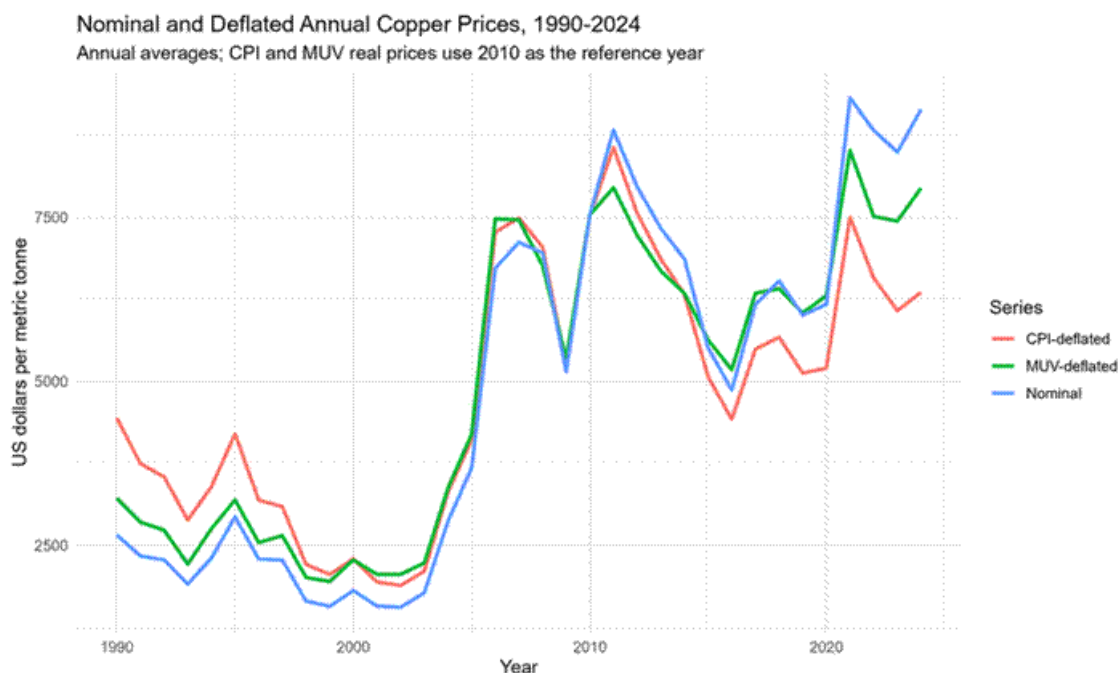


Figure 1 — Nominal vs deflated prices: overall, the nominal price shows higher oscillations than the deflated prices, indicating that copper price's non-stationarity is substantially influenced by manufacturer's side as well as consumer side inflation to varying degrees.

It can be seen in this graph that the vertical order of the trends becomes a mirror of itself at 2010 - before 2010, the nominal price is the lowest and CPI is the highest while MUV is in the middle - this is because 2010 has been

taken as the base year for the deflators - and they are lower than 100 before 2010. Since the formula for deflation is:

$$\text{Real copper price} = \text{Nominal copper price} \div (\text{Deflator in that year} \div \text{Deflator in 2010})$$

In 2010, the denominator became 1, hence the real price and nominal price meet. Before 2010, the denominator becomes less than 1, hence the real price can be seen to be greater than the nominal price - this is **actually not true in reality** but it is just a **function of what base year we have taken** - this simply means that **inflation was lower than the base year at that point**, hence it appears that real prices were lower than nominal prices - but after 2010, inflation was higher than the 2010 value, hence the real price becomes lower than the nominal price. This chart also shows how the choice of base year influences the visual output of the graph - and accurate interpretation and knowledge of the assumptions behind the graph is key to understanding copper price movement.

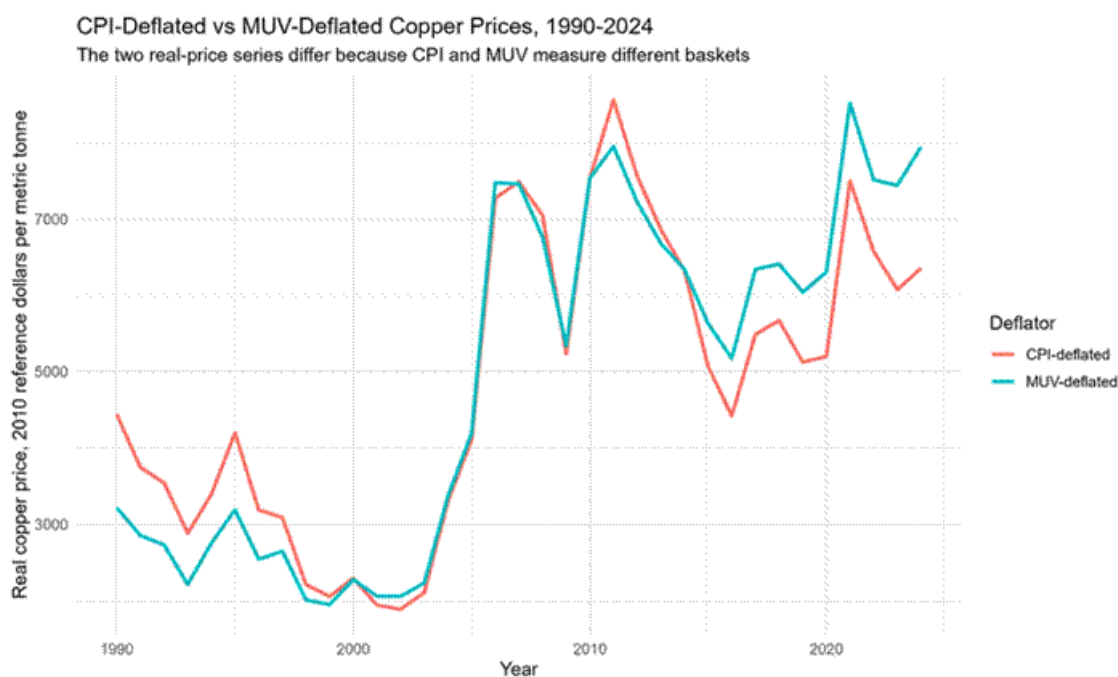


Figure 2 — CPI-real vs MUV-real: the two real price (deflated from nominal) lines side by side. MUV-real sits higher and rises more than CPI-real.

This can be interpreted as the fact that consumer prices in the USA have grown so much because of demand; and the purchasing power of a single dollar has shrunk so much over time that when the nominal price is deflated by CPI, the drop is far greater than when it is deflated with MUV.

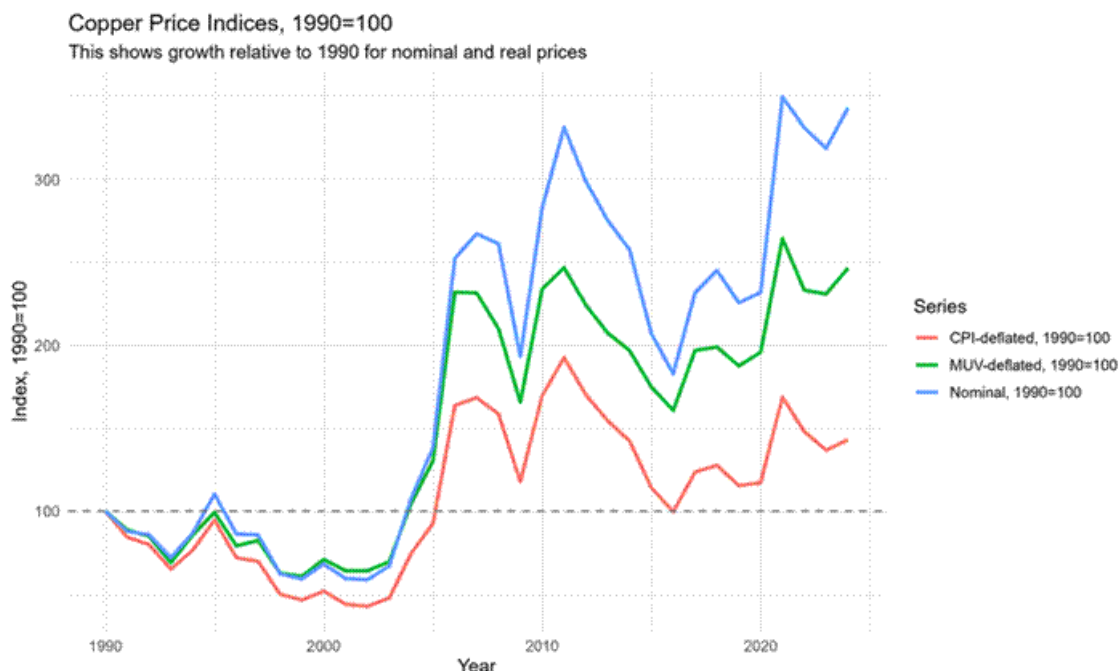


Figure 3 — Indices (1990 = 100): all three copper price trends have been compared against a **base level of 100 at 1990** to understand their relative growth and divergence.

By this we can infer that **manufactured exports have not grown in price as much as end-consumer goods** have - which is why the CPI deflated trend is much lower than the other two - since consumer prices have ballooned since 1990. The logic behind this is to observe the ratio in which these three trends have travelled together. The starting year's value is set to exactly 100, and then every year thenceforth is expressed as a percentage of the starting point. This makes different series comparable by assuming a common starting point. It is a method of standardisation. because they all begin from the same 100. The choice of baseline year is normative, and can be changed based on the desired objective.

****An important distinction to make here is that the 2010 dollar value reference made earlier is used for deflation:**

$\text{CPI-real price} = \text{nominal price} * \text{CPI in 2010} / \text{CPI in that year}$

$\text{MUV-real price} = \text{nominal price} * \text{MUV in 2010} / \text{MUV in that year}$

This converts nominal copper prices into 2010 price-level terms.

In this chart, **1990 = 100** is only used for **indexing** the already-computed series so the chart can **compare growth** from the same starting point.

$\text{Index value} = \text{value in year } t / \text{value in 1990} * 100$

The workflow behind this is:

Step 1: Nominal copper price is deflated using CPI or MUV, with 2010 as the real-price reference year.

Step 2: Nominal, CPI-real, and MUV-real series are re-indexed so 1990 = 100 for visual comparison.

The figure does not mean that the deflator base (i.e. the reference dollar value) is taken from 1990. It means that all three plotted lines start at 100 in 1990 so that relative growth can be compared (**normalisation**).

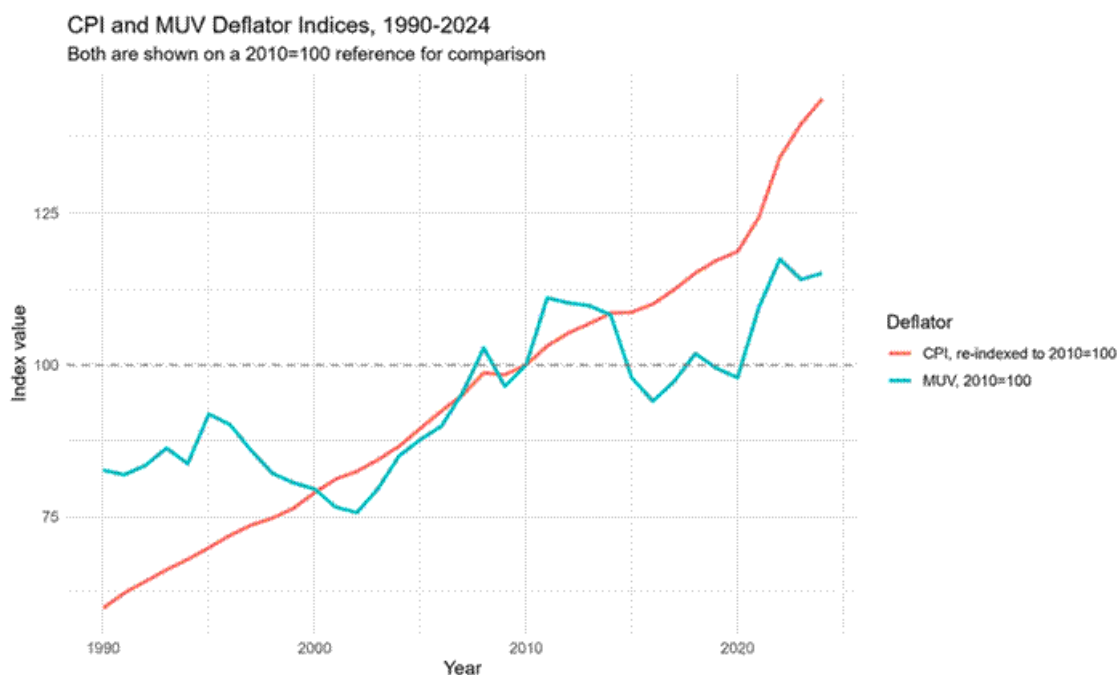


Figure 4 — Deflator indices, raw/actual values. MUV is seen to be far more erratic while CPI shows a homogeneously increasing trend with an even steeper rise after 2020.

Key trend increase values (1990 → 2024):

Nominal copper prices have increased from **\$2,661 → \$9,142** which is a net increase of **+243.5%**.

CPI-adjusted real copper prices have risen from **\$4,442 → \$6,355** which is a net increase of **+43.1%**;

MUV-adjusted real price went from **\$3,219 → \$7,943** which is a net increase of **+146.7%**.

This shows that a major part of the **nominal copper price rise** was caused solely by inflation. In real terms (inflation adjusted or deflated), copper prices still have risen, but they have risen much more strongly against global manufactured goods (whose prices themselves have not grown too much), and modestly against US consumer prices (which have genuinely ballooned, matching the huge oscillations of global metal prices).

2. Trend tests

A straight line was fitted to each series (price = intercept + slope × year). All three slopes are positive and statistically significant.

Series	Slope (\$/year)	p-value	R-squared
--------	-----------------	---------	-----------

Nominal	+221.5	2.5×10^{-10}	0.707
CPI-deflated (real)	+122.2	4.1×10^{-5}	0.404
MUV-deflated (real)	+182.1	1.2×10^{-9}	0.678

A rank-based **Spearman** check (less sensitive to erratic spikes which may have been caused by random market shocks) also verifies the result: $\rho = 0.76$ (nominal), 0.63 (CPI-real), 0.74 (MUV-real), all $p < 0.001$.

Interpretation: the rise in real copper prices is a genuine long-term drift that has been authenticated post inflation adjustment.

3. Stationarity and the ACF plots

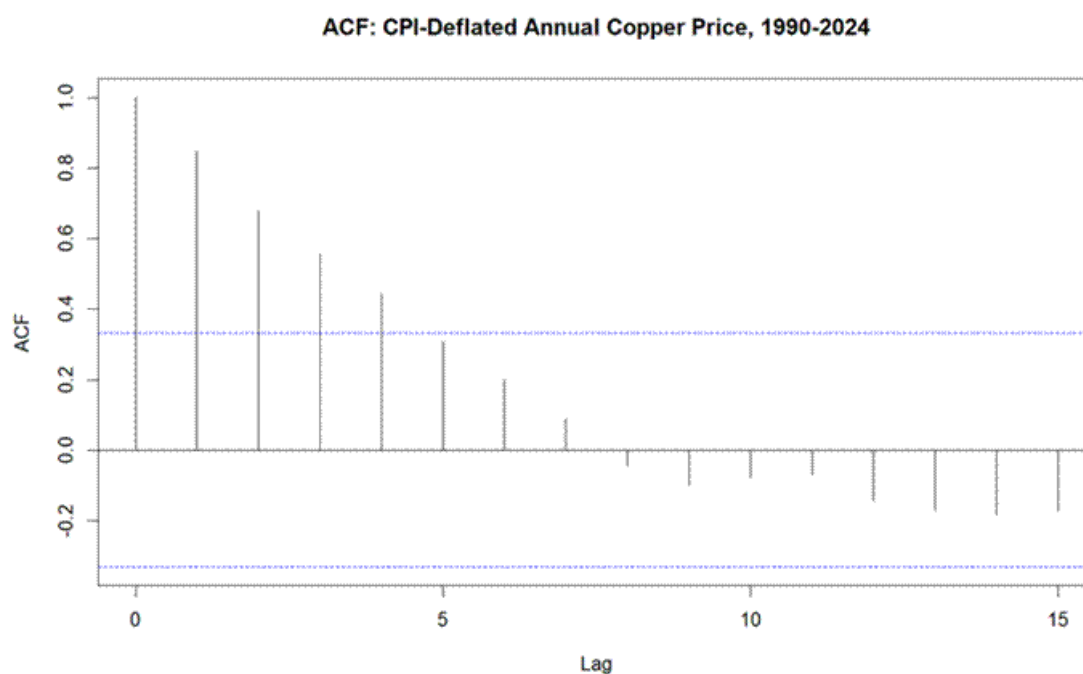


Figure 5 — ACF of the CPI-real level: bars stay high across several consecutive lags — the trend level has strong memory and wanders (non-stationary).

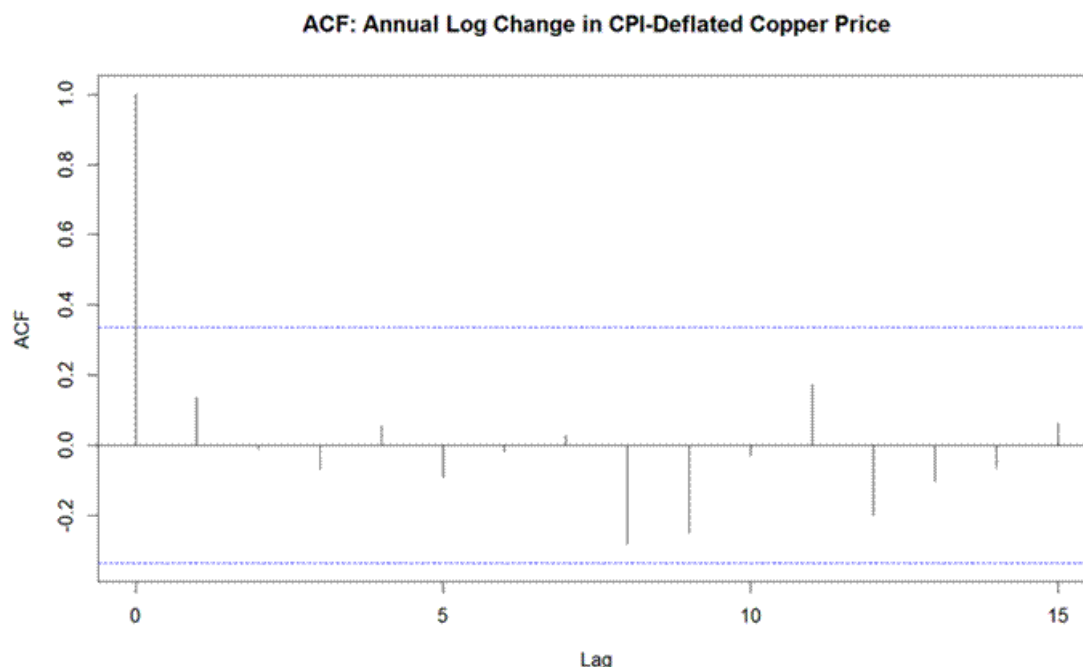


Figure 6 — ACF of year-on-year log change: bars drop to near zero almost immediately — the changes look much more stable.

Formal tests on CPI-real. Level: ADF $p = 0.654$ (cannot rule out non-stationarity) and KPSS $p = 0.032$ (rejects stationarity) — hence the series is non-stationary.

Log change: ADF $p = 0.452$, KPSS $p = 0.100$ — closer to stationary, but copper stays volatile.

Interpretation: the price levels cannot be treated as uniformly rising - the forecast must carry wide uncertainty, because that is the reality of the global copper market which is hyper-dependent on geopolitical tensions, intra and international policies, and supply chain disruptions.

4. Baseline forecasts to 2030

Auto ARIMA selection chose **ARIMA(0,1,0)** — **a random walk or naive model** — for all three series. A random walk gives a **flat** point forecast: assuming that no errors (random shocks) and no effects of autocorrelation are incorporated into the model, the best guess is that the last observed value is carried forward. The confidence interval widens every year since this is a highly exponential, definitively non-stationary series.

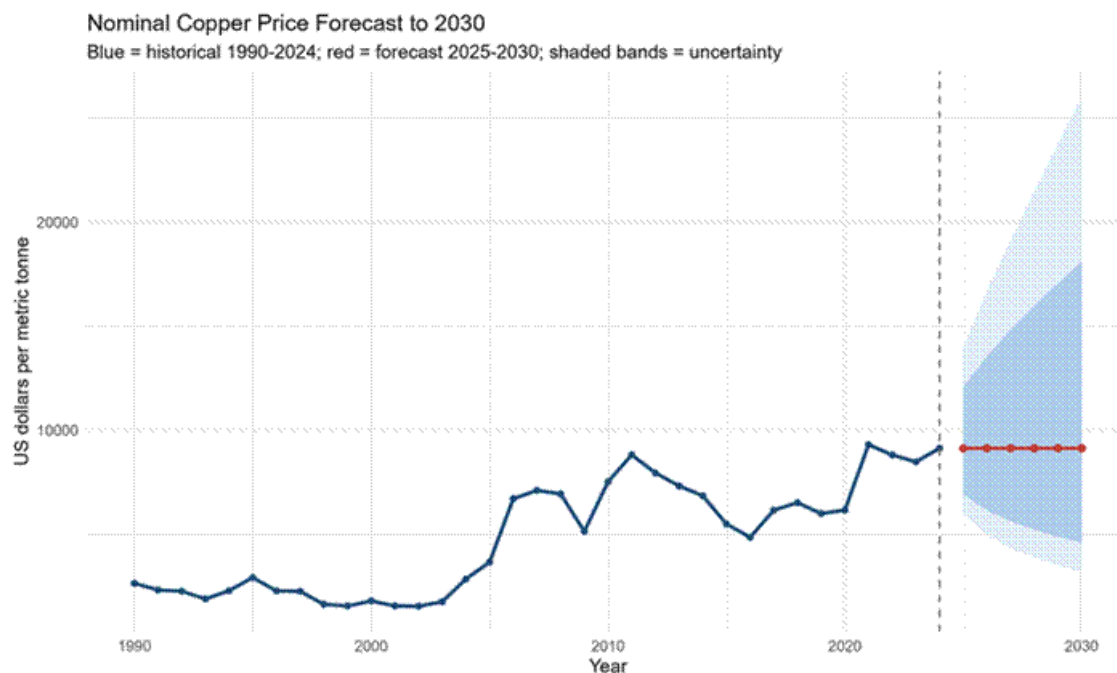


Figure 7 — nominal forecast: flat at \$9,142 to 2030; the 2030 95% range runs roughly \$3,213–\$26,010 which is a very wide uncertainty band.

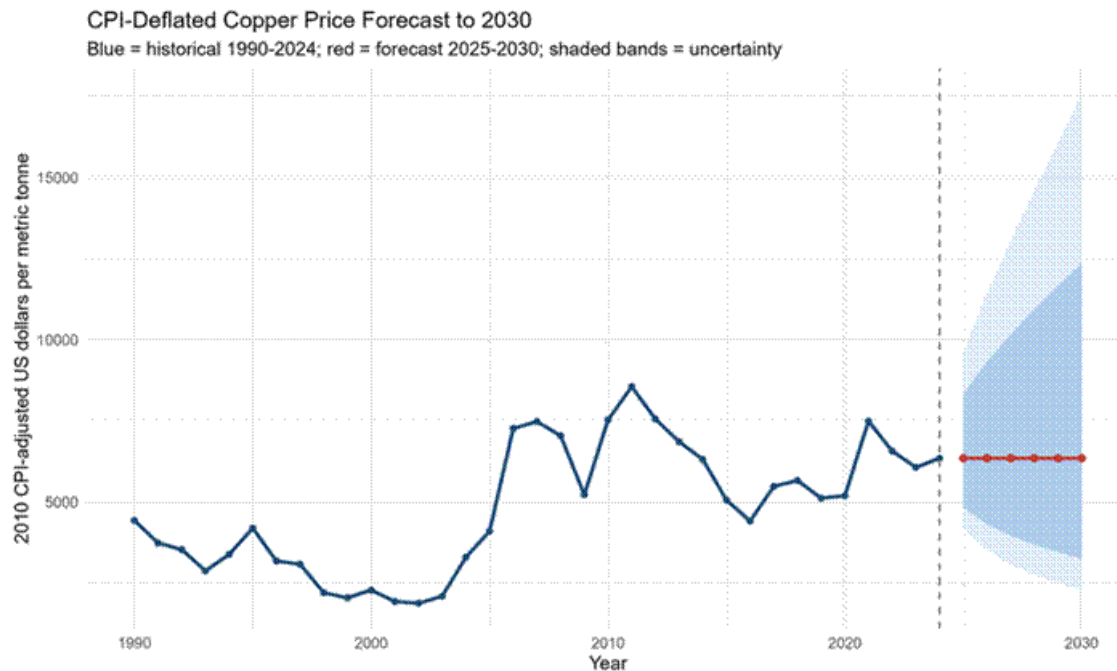


Figure 8 — CPI-real forecast: flat at \$6,355 to 2030, with a similarly wide fan.

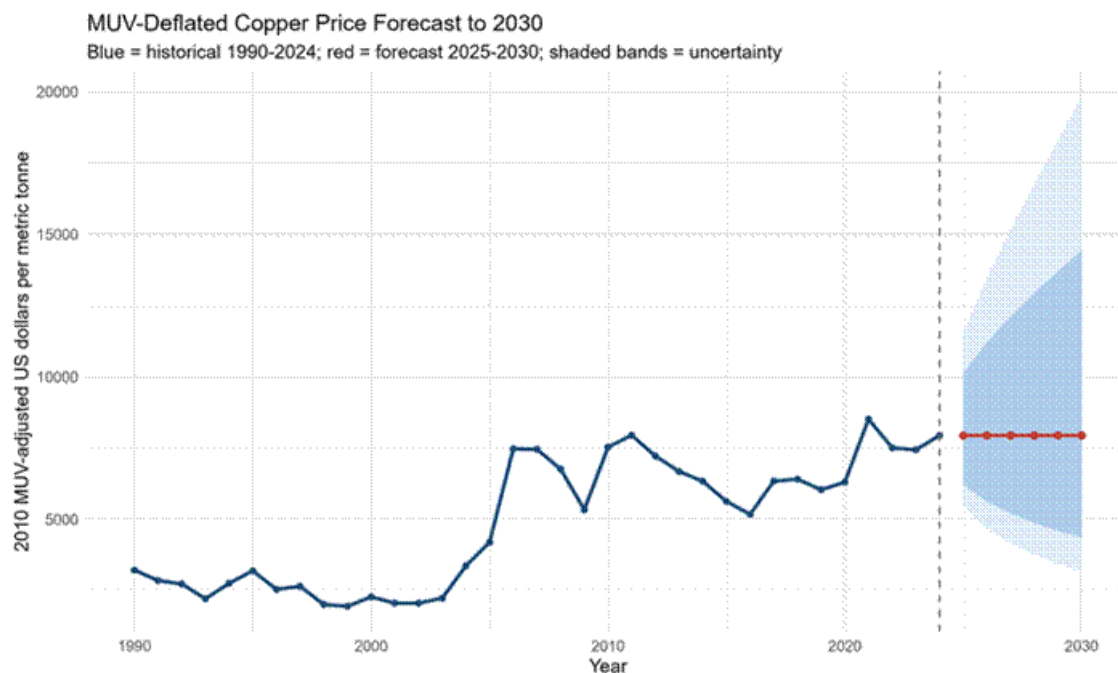


Figure 9 — MUV-real forecast: flat at \$7,943 to 2030.

Interpretation: the flat line is a conservative baseline, but the reality is that the future could vary dramatically.

5. Scenarios and validation

Because one flat line hides possible future fluctuations, three variations with some inherent movement were added: **Historical drift** (continues average past growth), **Momentum AR(1)** (lets recent changes influence the future path), and **Shock smoothing MA(1)** (dampens the effects of outlier shocks).

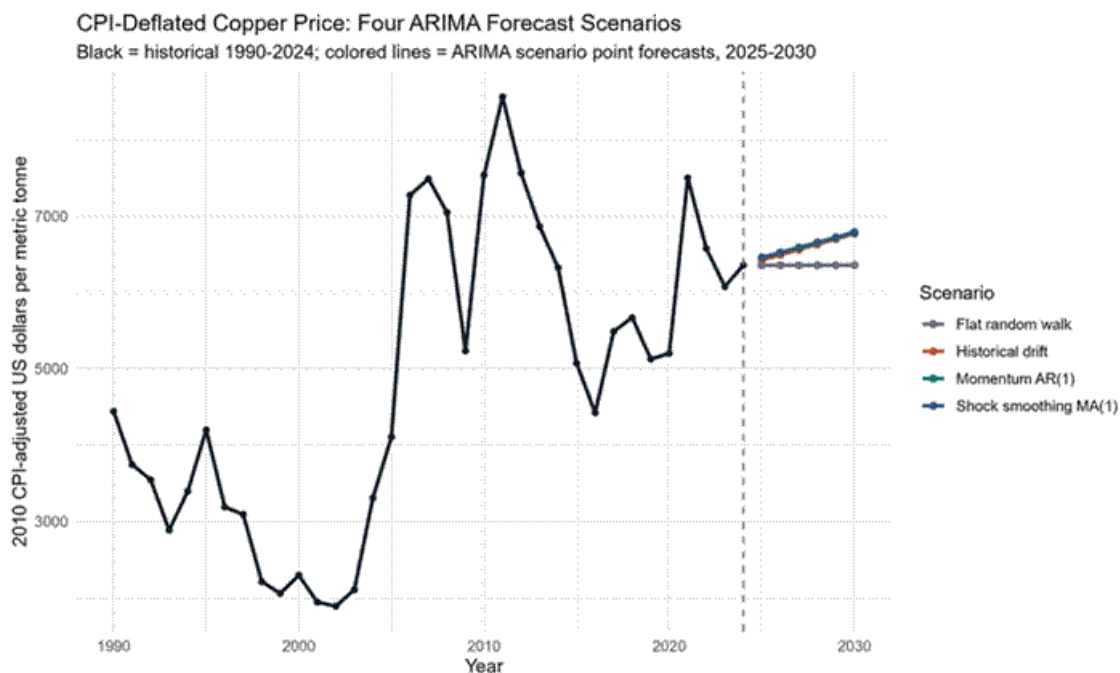


Figure 10 — CPI-deflated real price, four scenarios: the flat path plus the three moving paths to 2030.

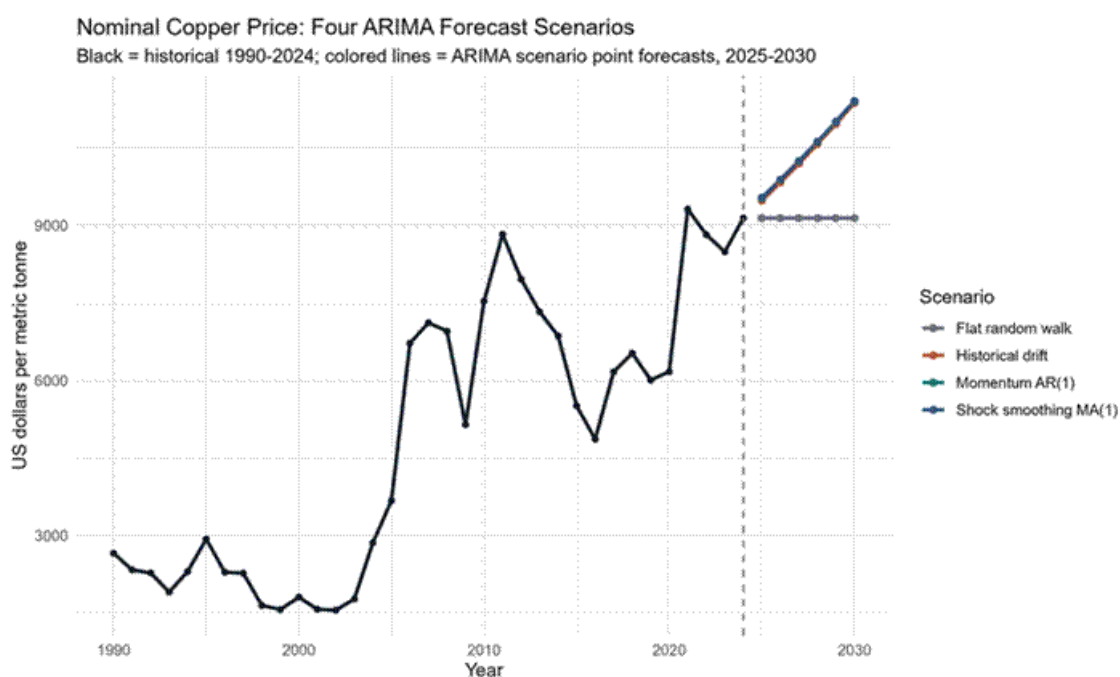


Figure 11 — nominal price, four scenarios: same four scenarios on the nominal series.

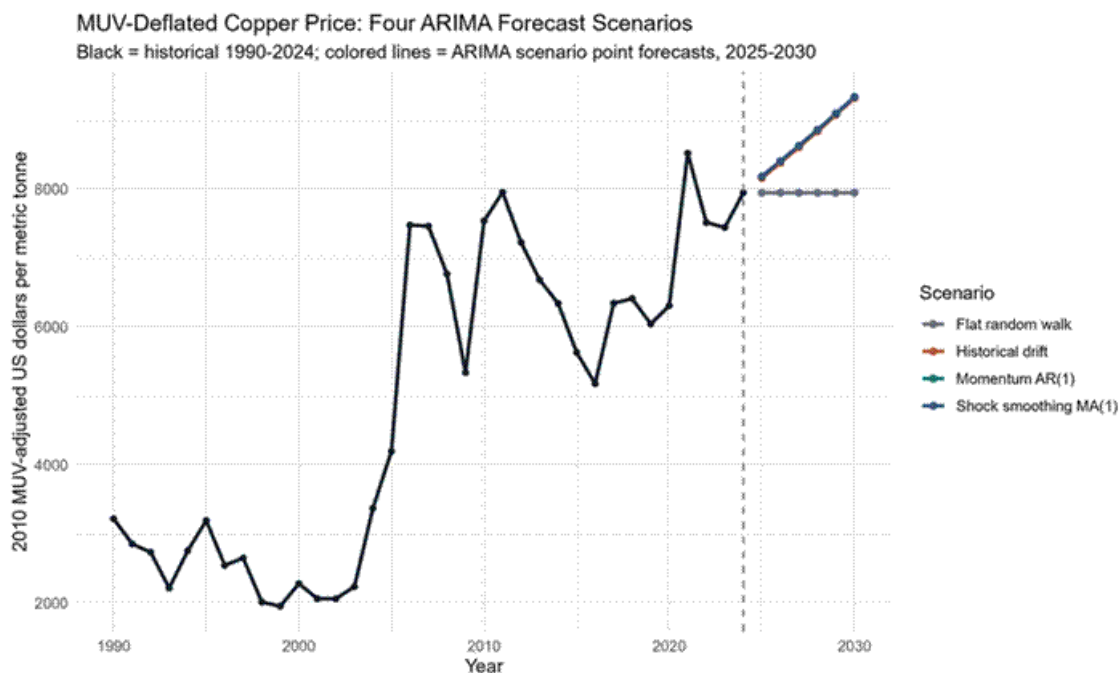


Figure 12 — MUV-real, four scenarios: same four scenarios on the MUV-real series.

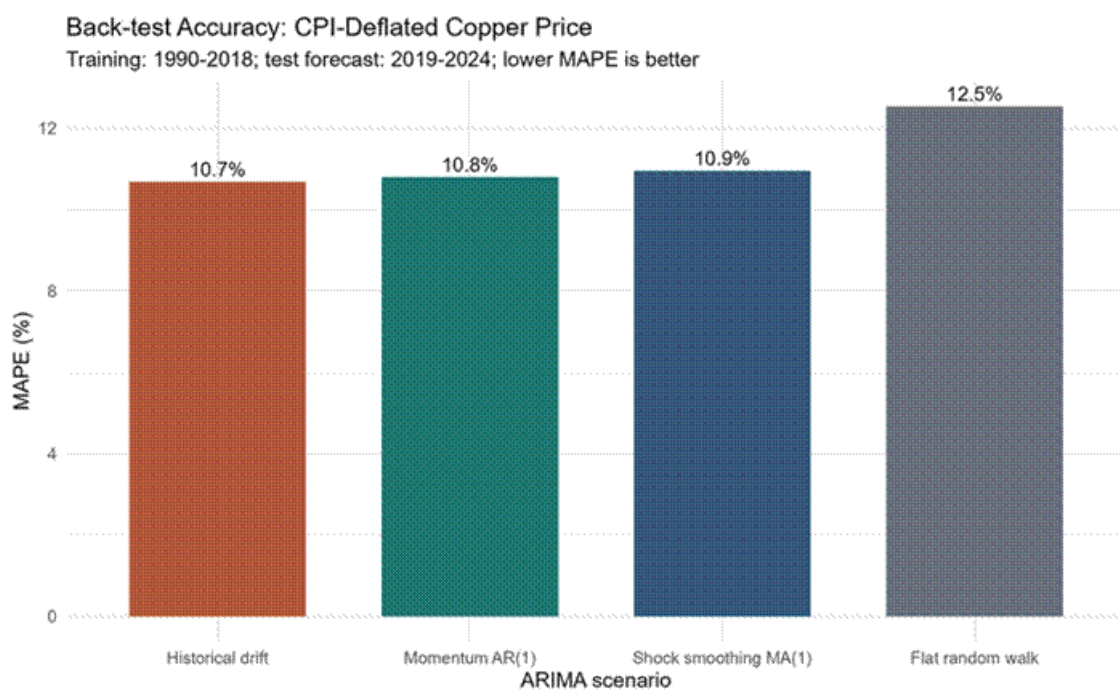


Figure 13 — back-test (validation) (MAPE): each model was trained on 1990–2018 and scored on the 2019–2024 hold-out.

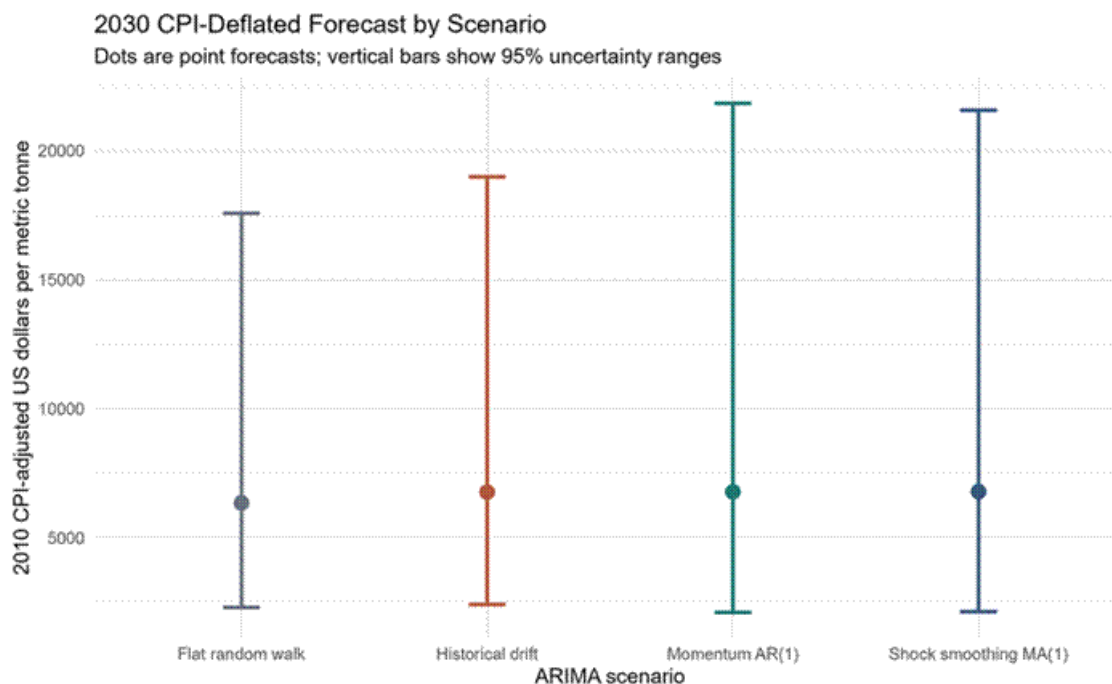


Figure 14 — 2030 ranges: the CPI-deflated real prices forecasted to 2030 with their wide intervals.

Scenario (CPI-real)	2030 point (\$)	Back-test error (MAPE)
Flat random walk (naive)	6,355	12.5%
Historical drift	6,770	10.7% (most accurate)
Momentum AR(1)	6,780	10.8%
Shock smoothing MA(1)	6,789	10.9%

All three moving scenarios have better accuracy than the naive model since they capture copper prices' upward movement better. But the four 2030 points (\$6,355–\$6,789) are close to one another while their 95% ranges (about \$2,100–\$21,800) overlap heavily.

Interpretation: The scenarios have similar prediction powers, but their wide uncertainties show that no one number can definitely be quoted for the copper price of any given year in the future - only a range of values can be given with the price tentatively falling within that range.

6. Regression: what explains nominal price?

What percentage of nominal copper-price movement can CPI and MUV explain?

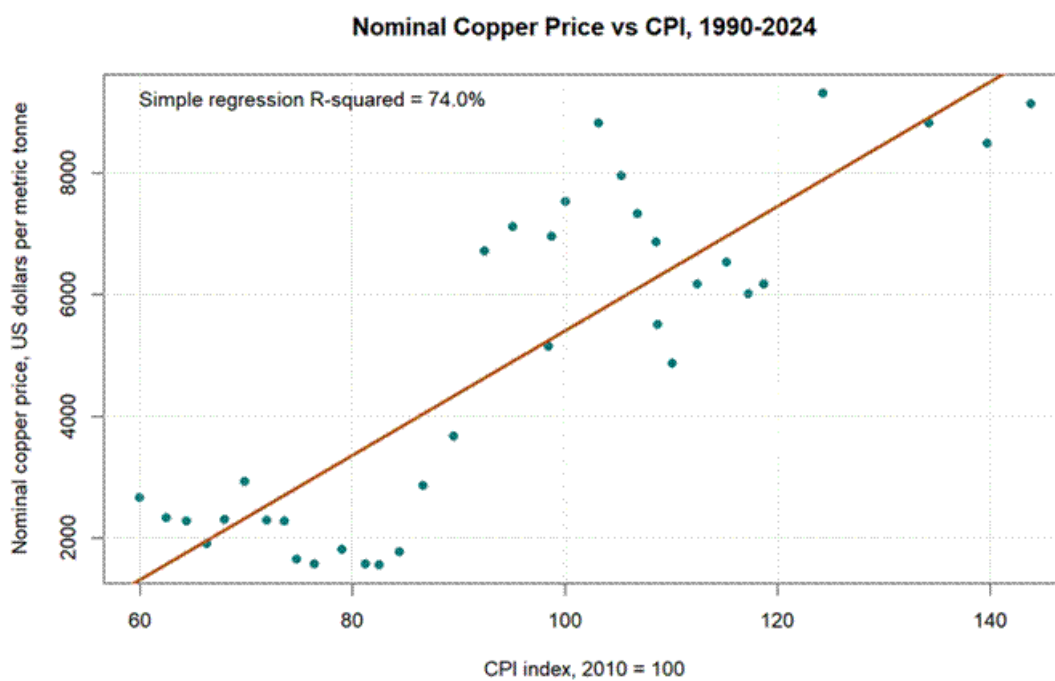


Figure 15 — nominal vs CPI: scatter plus fitted line; CPI alone explains 74.0%.

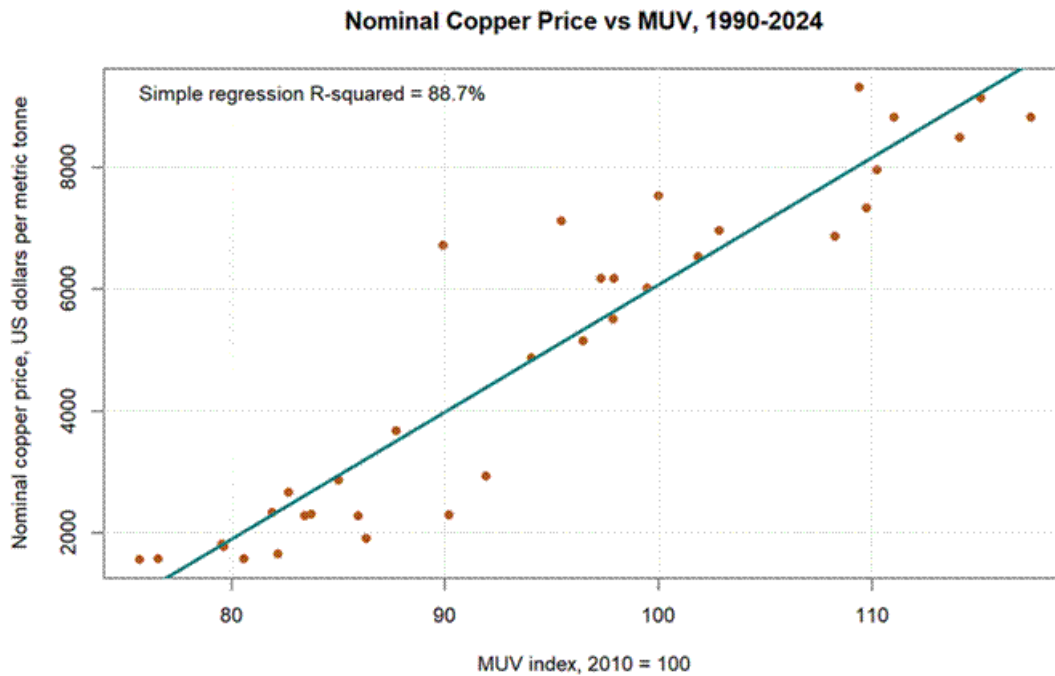


Figure 16 — nominal vs MUV: a tighter fit; MUV alone explains 88.7%.

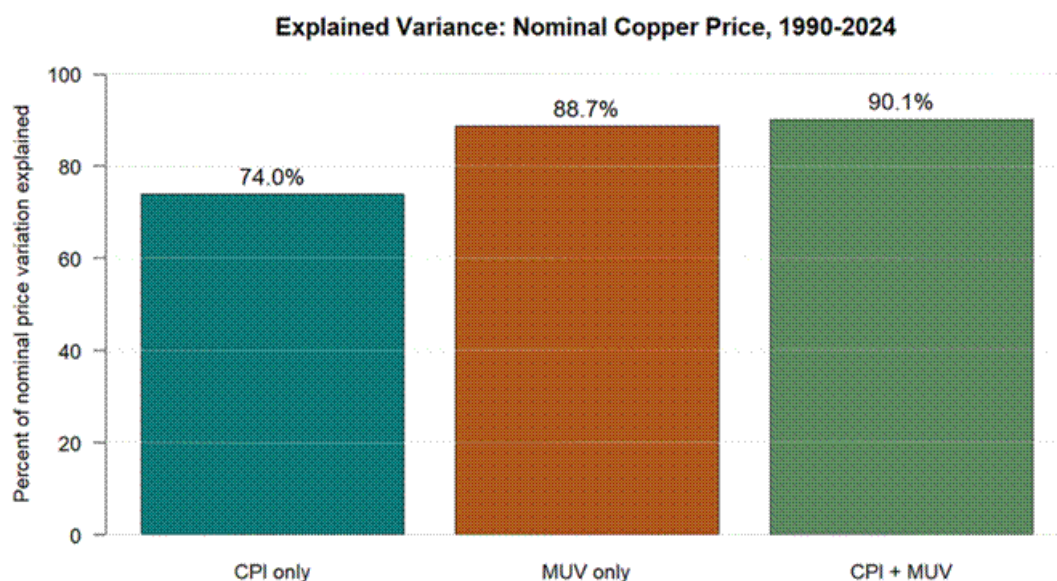


Figure 17 — explained variance: CPI 74.0%, MUV 88.7%, both together 90.1%.

An important point to note here is that correlation tests were done between CPI and MUV to see if they move together - since this can bias the R-squared of the multiple regression.

CPI vs MUV Pearson correlation = 0.8447

p-value = 1.78e-10

CPI vs MUV Spearman correlation = 0.7994

p-value = 8.45e-09

This shows that CPI and MUV are quite **strongly correlated**.

For multiple regression, the predictors should not be highly correlated since this creates multicollinearity. Multicollinearity makes it harder to interpret the separate independent effects of CPI and MUV - however, in this case the independent regressions have also been performed, so we clearly know the effect of each deflator on the nominal copper prices.

Hence, the **Variation Inflation Factor (VIF)** test was also done:

CPI VIF = 3.49

MUV VIF = 3.49

Interpretation: Since VIF can go from 1 to infinity, 3.49 shows that there is **moderate multicollinearity**, but not extreme (*10.7 - Detecting Multicollinearity Using Variance Inflation Factors | STAT 462, n.d.*). It does not make the multiple regression invalid, but it means that the individual CPI and MUV coefficients are more reliable than the combined one - and MUV has a higher coefficient of determination for copper prices than CPI.

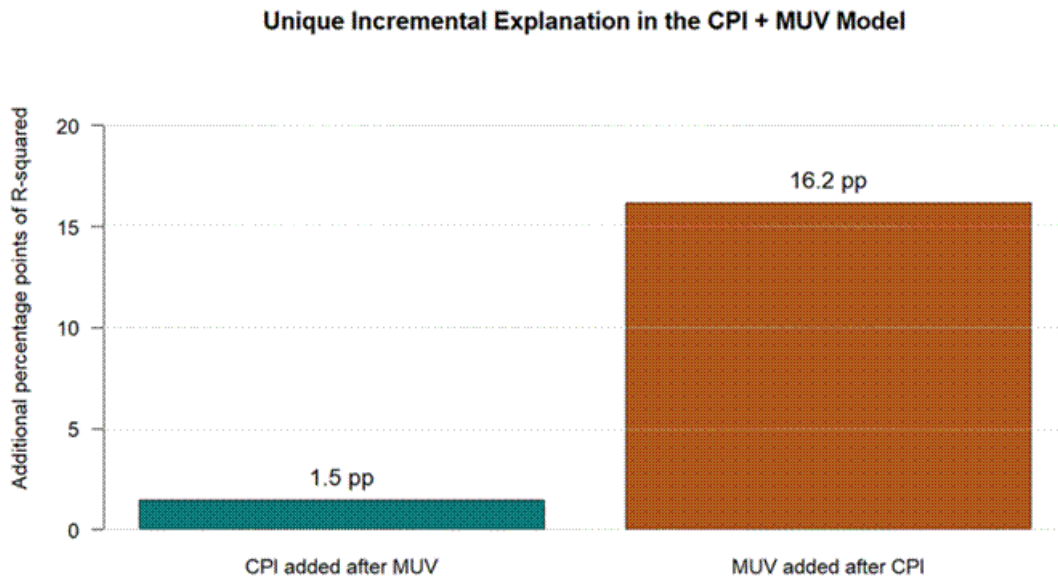


Figure 18 — Incremental: adding MUV on top of CPI lifts R-squared by 16.2 points; adding CPI on top of MUV lifts it only 1.5 — hence MUV has greater explanatory strength.

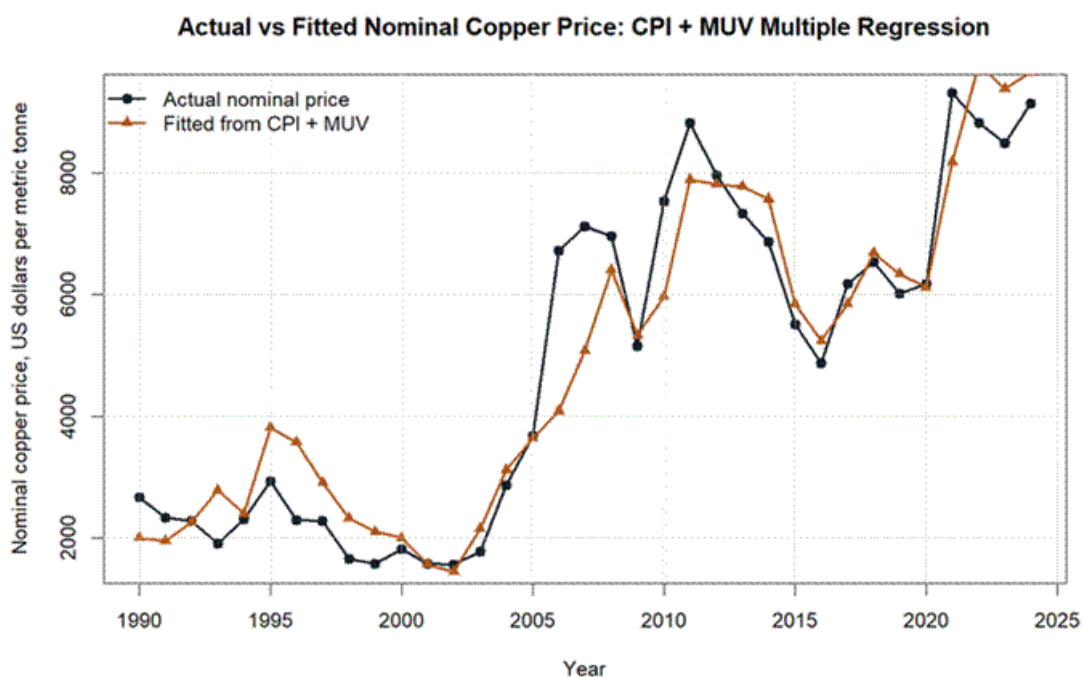


Figure 19 — actual vs fitted: the model with CPI and MUV together predicts the movement of the real values closely.

Coefficients (of the combined CPI-MUV model):

Intercept −13,370.8;

CPI +26.82 ($p = 0.037$);

MUV +166.55 ($p = 3.2 \times 10^{-8}$).

On a common scale the weights are 0.23 (CPI) vs 0.75 (MUV).

VIF = 3.49 (moderate overlap).

Caveat: These findings indicate **association, not causation**.

7. Correlation

The link between each deflator and nominal price.

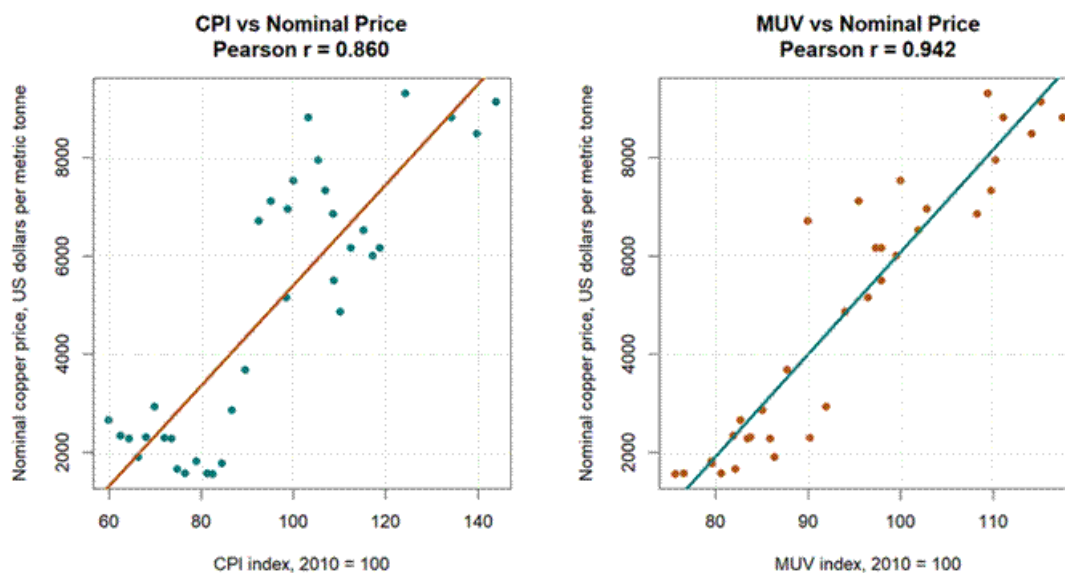


Figure 20 — scatter plot: nominal price against CPI and against MUV.

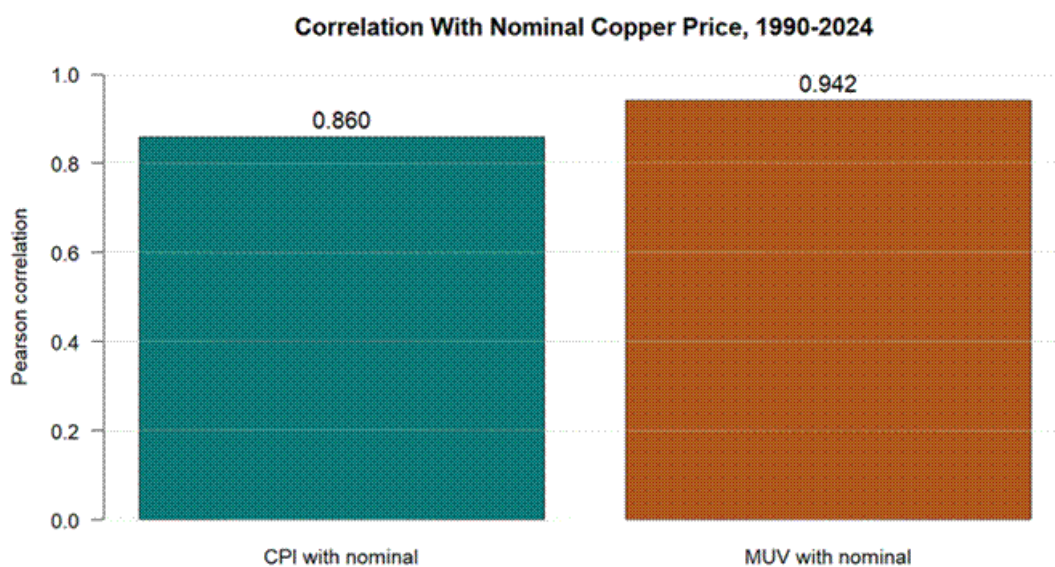


Figure 21 — bar chart: Pearson CPI 0.860, MUV 0.942; Spearman CPI 0.764, MUV 0.935.

Interpretation: Under both methods MUV is more strongly linked to nominal copper price than CPI is to the nominal price — confirming the findings of the regression analysis and emphasizing association rather than causation.

Appendix — USA Copper Records Distribution Maps & Charts

These maps show the geography (locations) of copper-related records for mining, prospects and occurrences, processing plants, and porphyry deposits (USGS MRDS, 2016 edition). “Grade A/B” is classified by the MRDS as ranking for data completeness of the records and do not indicate ore quality. “Oxide/sulfide” means which mineral terms are mentioned, not pure ore type.

**All maps are also attached as a zip folder along with the PDF report submission to enable individual inspection and zooming in, since the PNGs pasted here are blurry in some cases. All data are sources mentioned in the adjoining references section.

National distribution

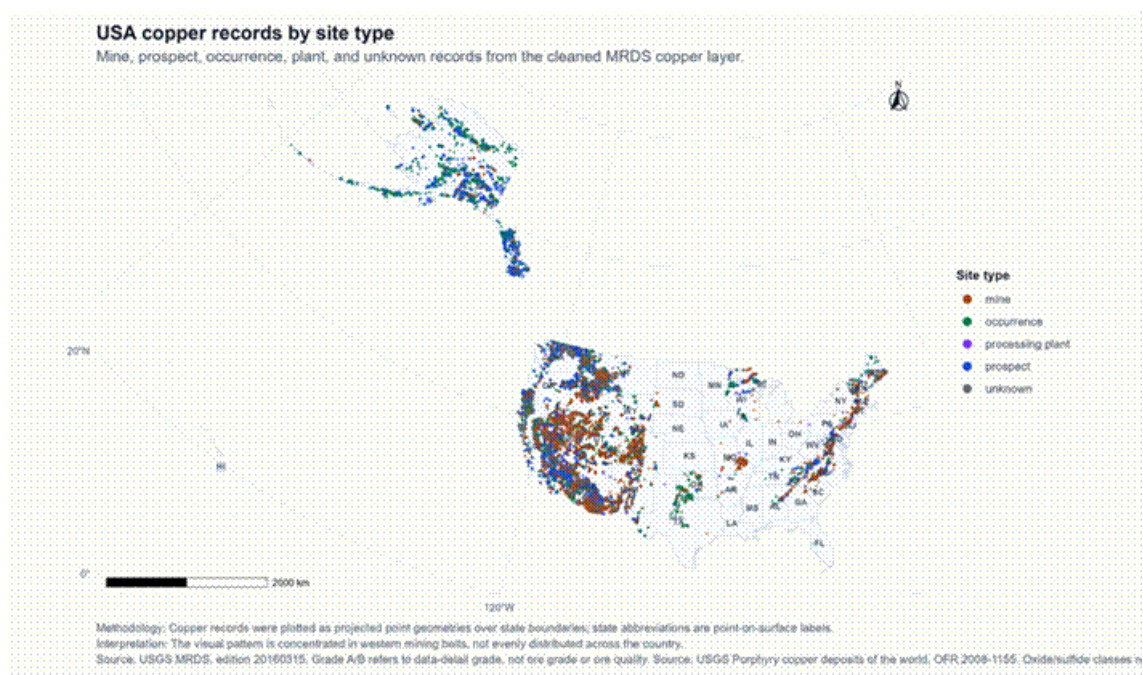


Figure 22 — National Point Map: every record coloured by type. Records cluster heavily in the western mining belts, Alaska, and a thin eastern strip — not spread evenly.

Methodology: Each cleaned USGS MRDS (2016 edition) copper record plotted as one projected point over US state boundaries, coloured by its site-type field.

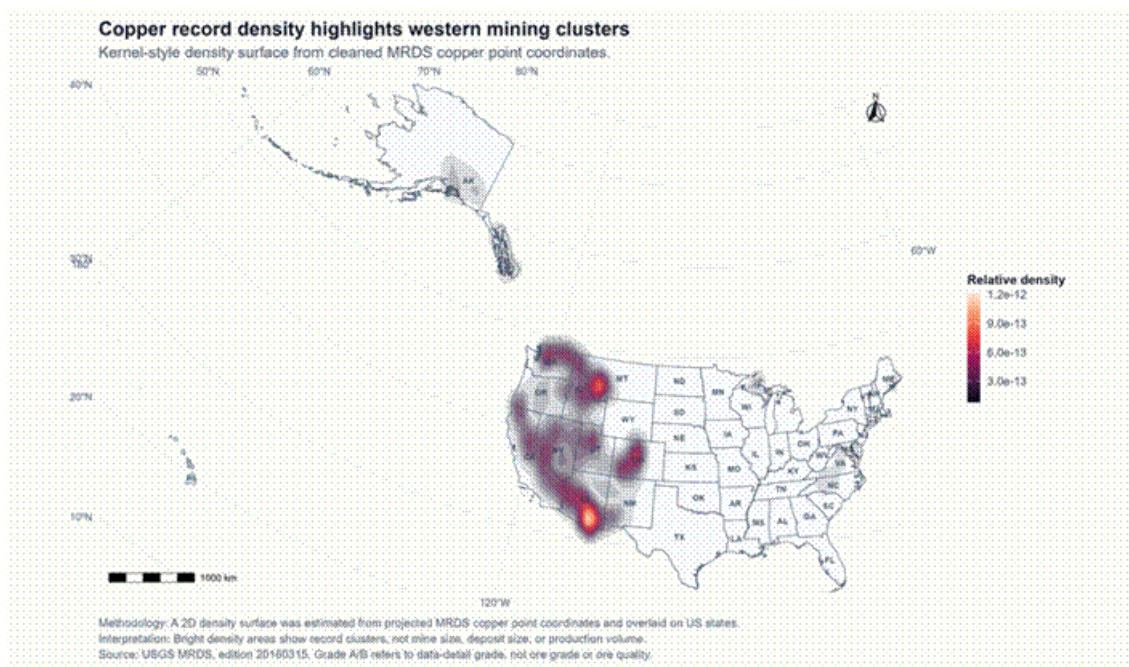


Figure 23 — Density Heatmap: bright zones mark where records cluster most densely per unit area.

Methodology: A kernel (2D) density surface estimated from the point coordinates

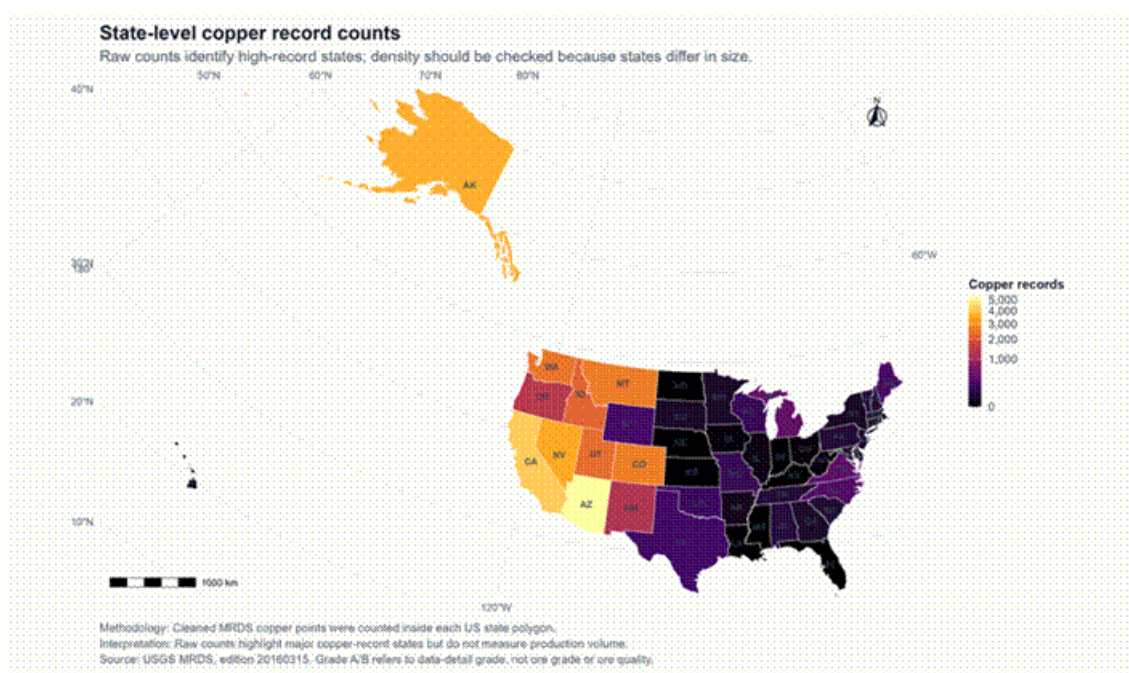


Figure 24 — State Choropleth: states shaded by total record count, using a square-root colour scale.

Methodology: MRDS copper points counted inside each state polygon

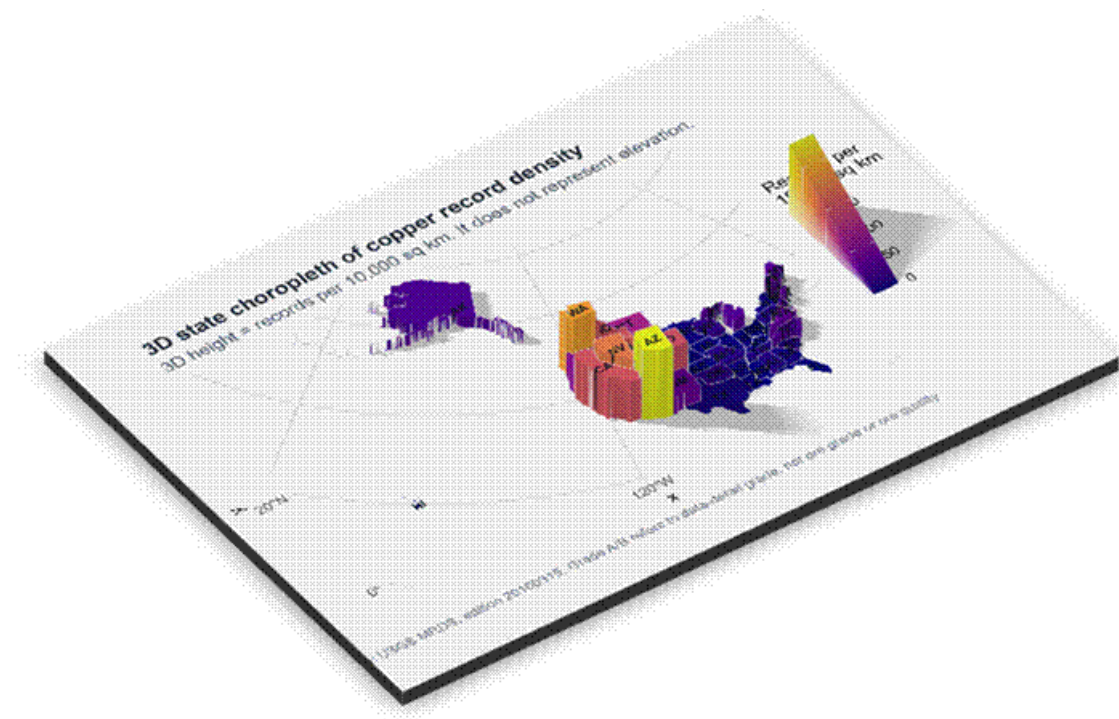


Figure 25 — 3D State-wise Density: record density shown as records per 10,000 square kilometers shown as 3D surface height combined with colours — height plus lighter colour indicates more records per area.

Methodology: Per-state record density extruded into 3D surfaces.

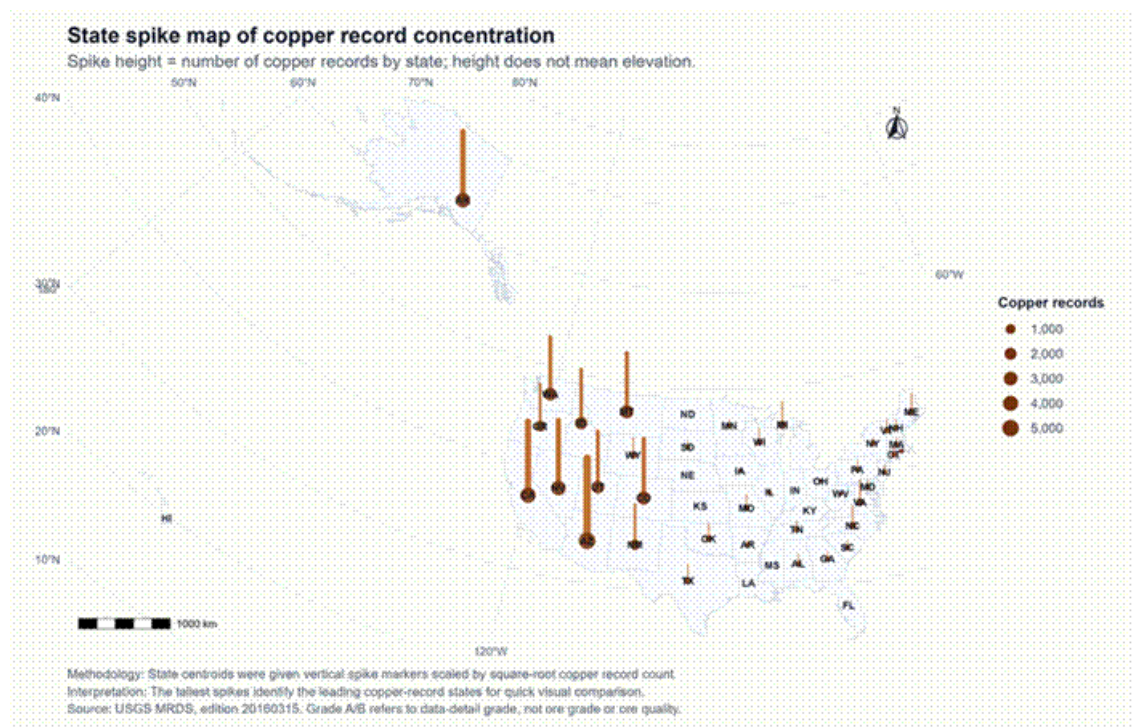


Figure 26 — Spike Map: state-wise record counts as vertical spikes — taller spike = more records

Methodology: The per-state record counts combined into vertical spikes rising from each state's location.

Ranking and Composition

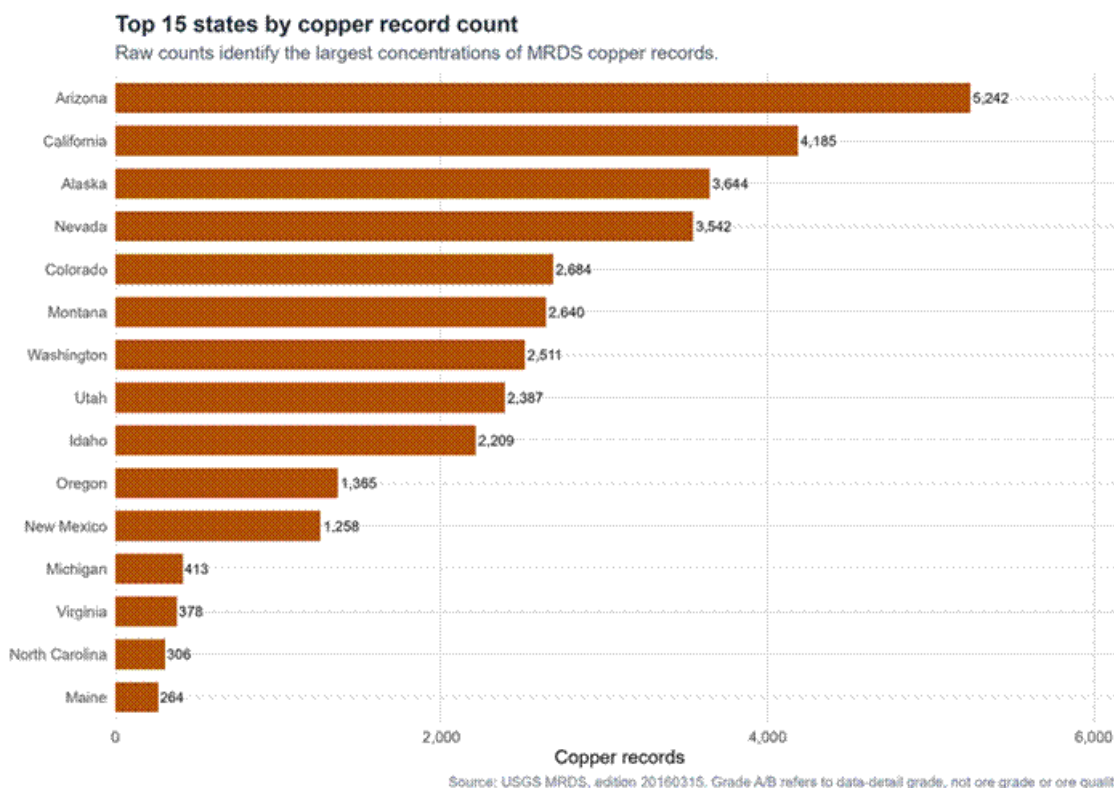


Figure 27 — Top 15 States: Arizona leads with 5,242 records, then California 4,185, Alaska 3,644, Nevada 3,542, Colorado 2,684, Montana 2,640, Washington 2,511, Utah 2,387, Idaho 2,209, Oregon 1,365, New Mexico 1,258. Eastern states trail far behind (Michigan 413, Virginia 378, North Carolina 306, Maine 264).

Methodology: MRDS copper records tallied per state and the 15 highest counts ranked into a horizontal bar chart.

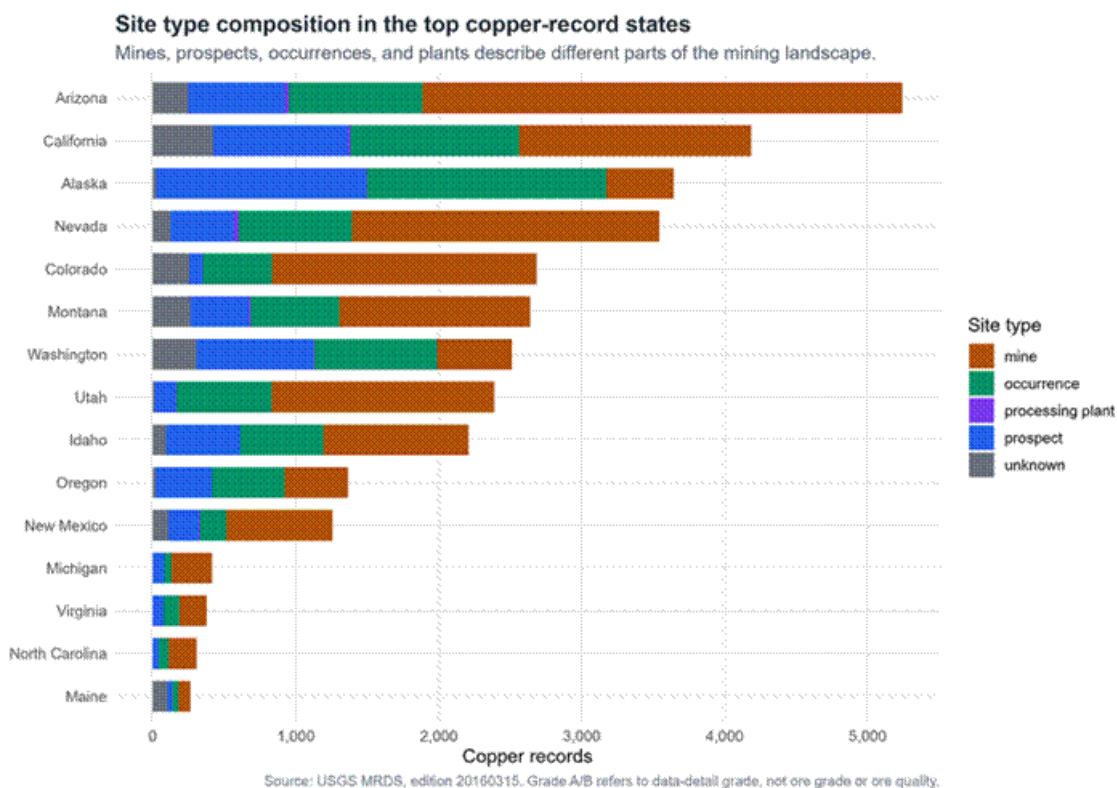


Figure 28 — Site-Type Composition: the top states split into mine / prospect / occurrence / plant / unknown. Arizona and Nevada are dominated by mines; Alaska has more prospects and occurrences — more exploration-stage than developed.

Methodology: For those top states, the same records are split by their site-type field and stacked to show each state's mix.

By site type and data grade



Figure 29 — Mines: the developed, historical mining geography.

Methodology: The MRDS copper layer filtered to records flagged "mine" and mapped as points.

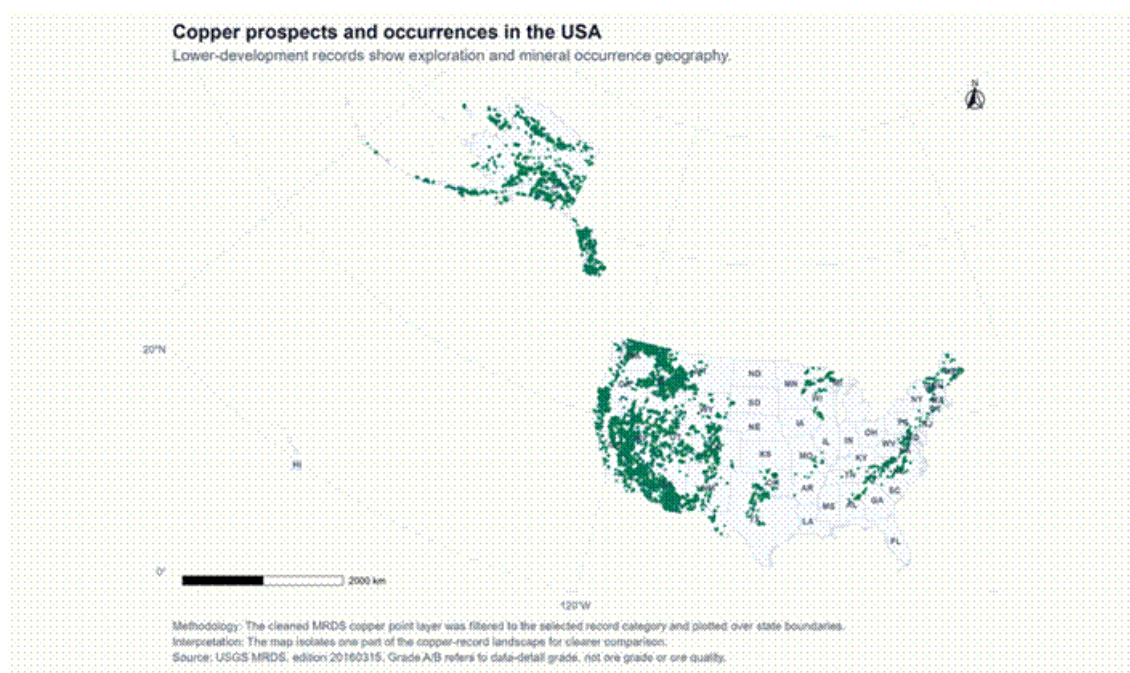


Figure 30 — Prospects & Occurrences: the geography of future exploration.

Methodology: The MRDS copper layer filtered to "prospect" and "occurrence" records and mapped as points.

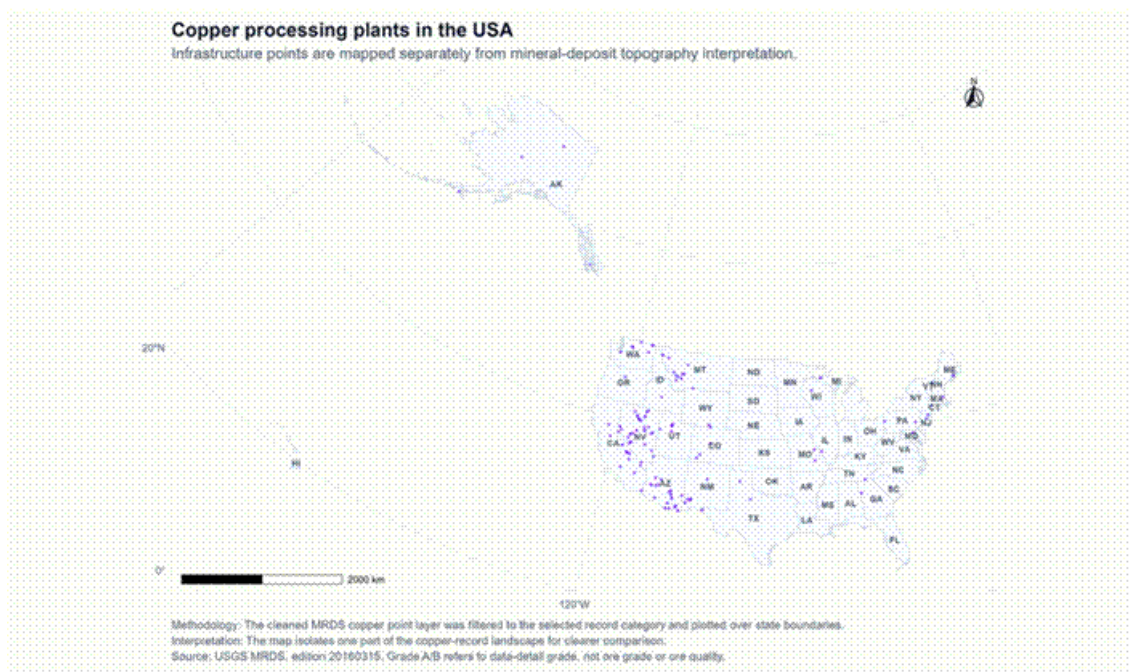


Figure 31 — Processing Plants: critical infrastructure for the copper supply chain.

Methodology: The MRDS copper layer filtered to "processing plant" records and mapped as points.

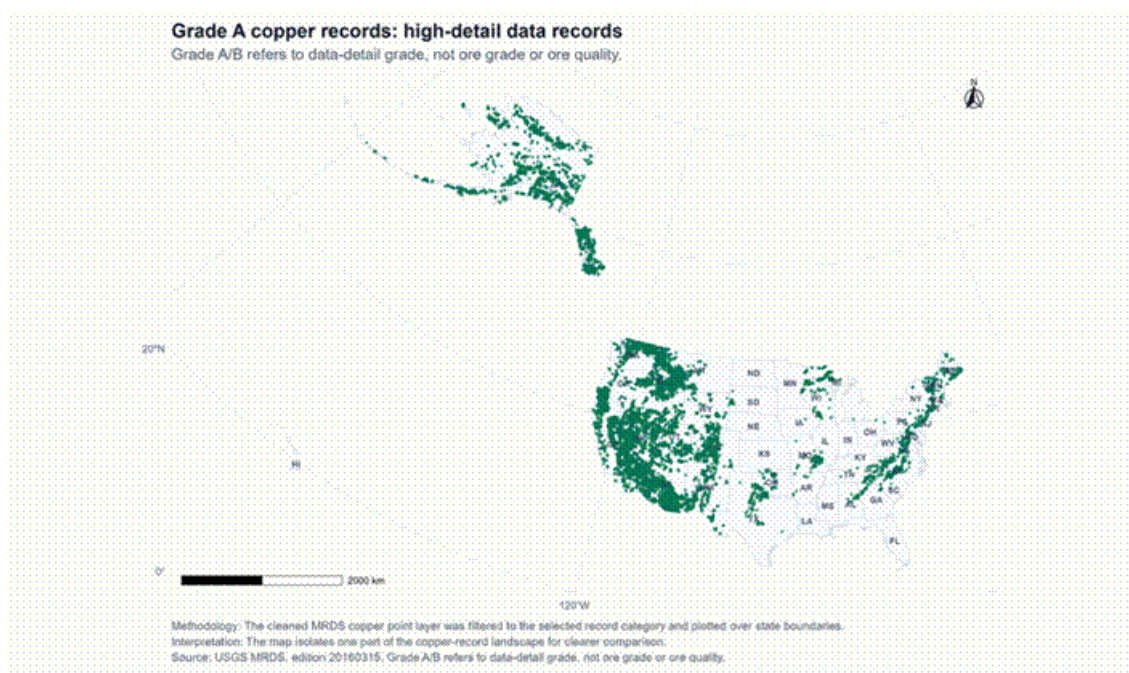


Figure 32 — Grade A records: highest data-completeness records.

Methodology: Records filtered to the higher data-completeness ("Grade A") class and mapped — based on record detail, not ore quality.

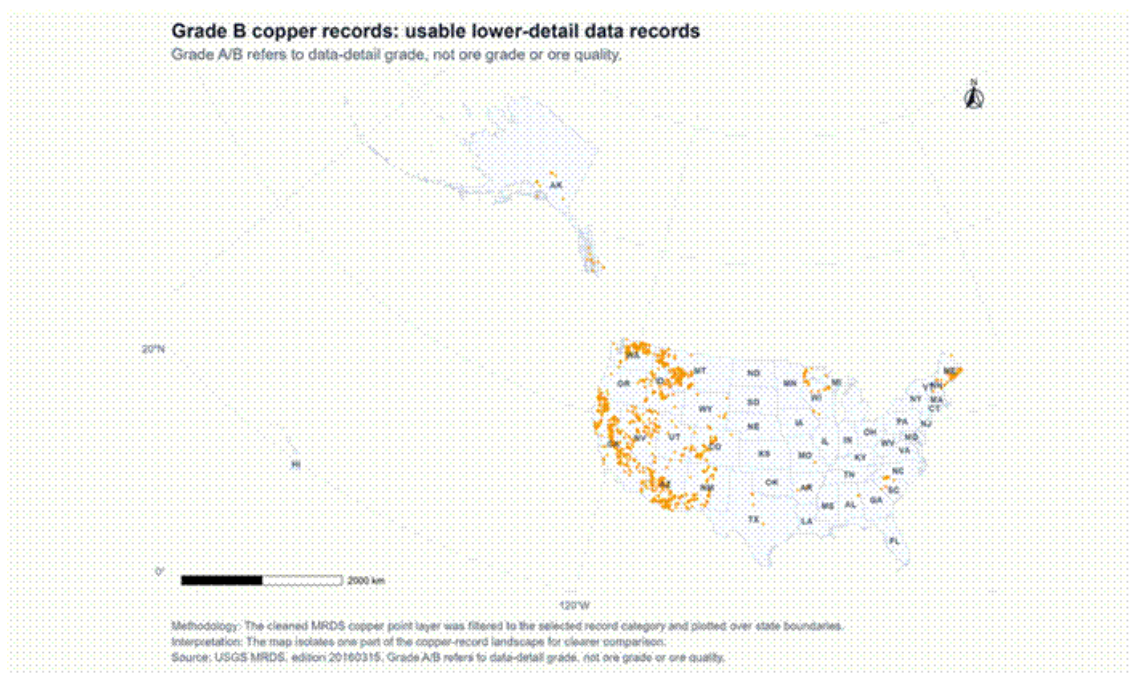


Figure 33 — Grade B records: lower data-completeness records (still usable).

Methodology: Records filtered to the lower data-completeness ("Grade B") class and mapped on the same basis.

Porphyry mineralogy and composite

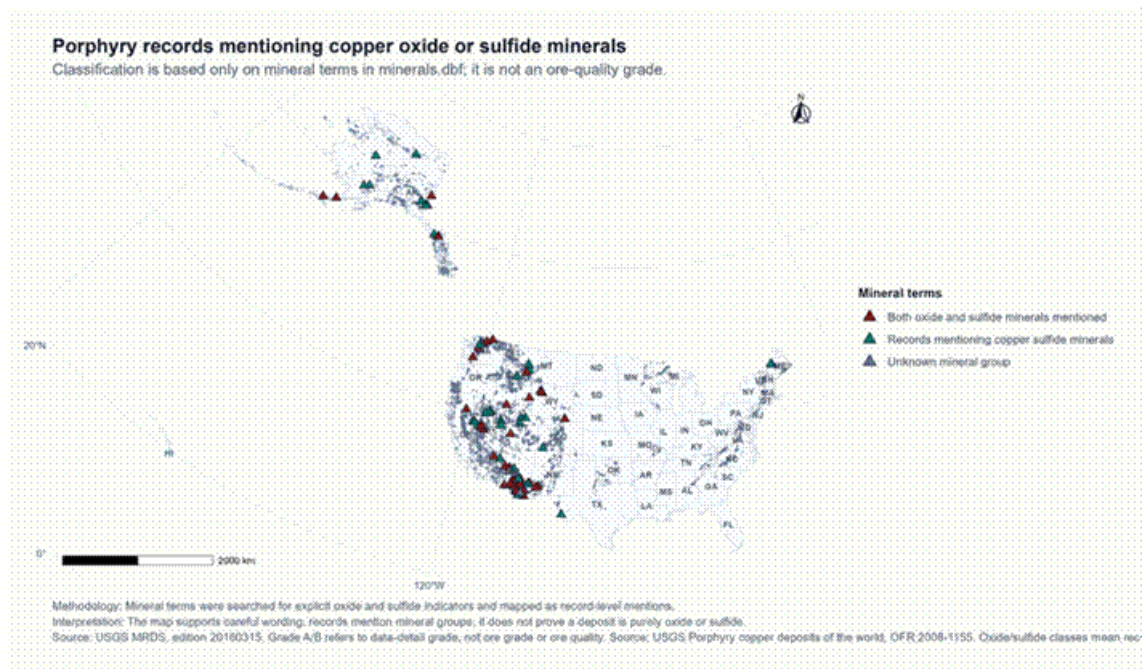


Figure 34 — Oxide/Sulfide Map: porphyry records flagged by whether oxide, sulfide, or both mineral terms appear (not pure ore type).

Methodology: Porphyry copper deposits (USGS OFR 2008-1155) classified by whether their minerals.dbf terms mention oxide, sulfide, or both, then mapped as points.

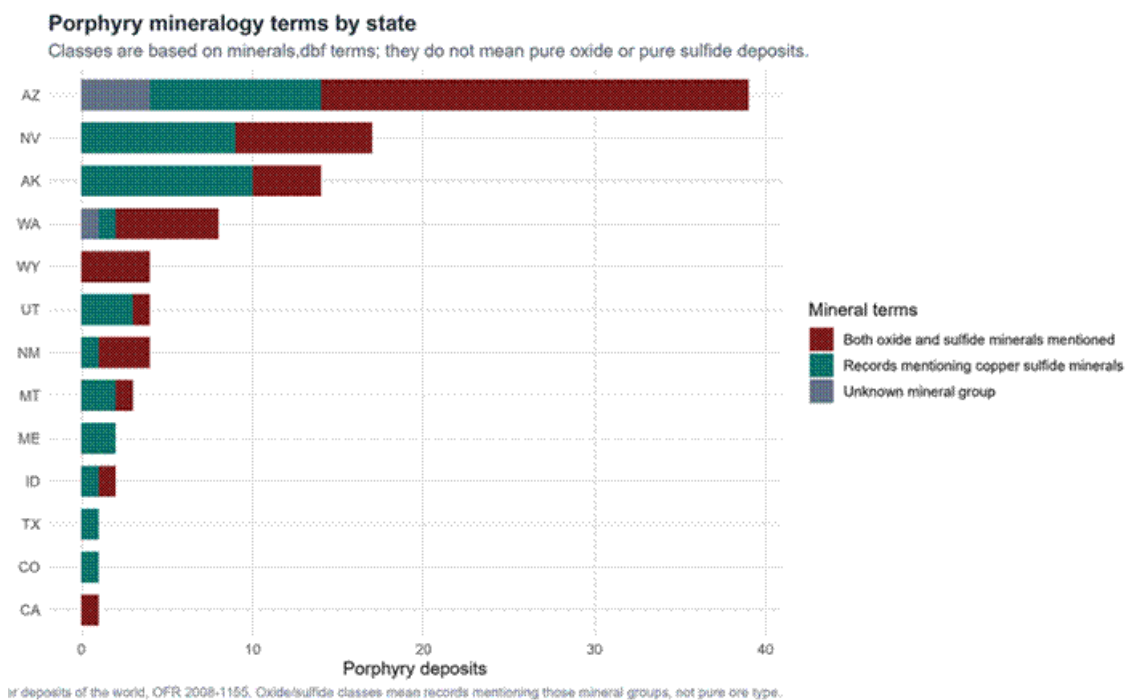


Figure 35 — Mineralogy by state: Arizona dominates porphyry records (about 39), then Nevada (about 17) and Alaska (about 14); most fall in the “sulfide” or “both” classes.

Methodology: Those oxide/sulfide/both porphyry classes counted per state and shown as stacked bars.

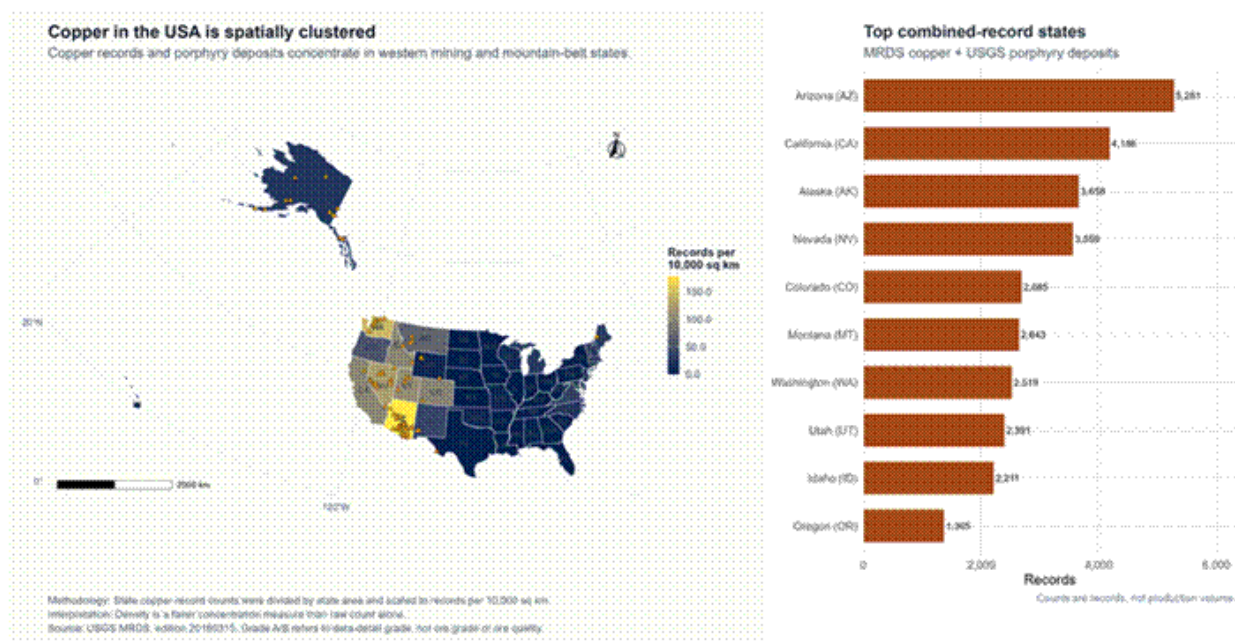


Figure 36 — Story composite: a summary tying the national pattern together

Methodology: The national point pattern and the top-state ranking combined into a single summary panel.

References for Copper Price and Spatial Visualisation

Alderman, A., Camp, K. M., & Mandiak, E. (2023, March). How currency appreciation can impact prices: The rise of the U.S. dollar. *Beyond the Numbers*, 12(6). U.S. Bureau of Labor Statistics.

<https://www.bls.gov/opub/btn/volume-12/how-currency-appreciation-can-impact-prices-the-rise-of-the-us-dollar.htm>

Cuddington, J. T., & Jerrett, D. (2008). Super cycles in real metals prices? *IMF Staff Papers*, 55(4), 541–565.

<https://doi.org/10.1057/imfsp.2008.19>

Grilli, E. R., & Yang, M. C. (1988). Primary commodity prices, manufactured goods prices, and the terms of trade of developing countries: What the long run shows. *The World Bank Economic Review*, 2(1), 1–47.

<https://doi.org/10.1093/wber/2.1.1>

- Investopedia. (n.d.). *Group of Five (G5)*. Retrieved June 13, 2026, from <https://www.investopedia.com/terms/g/g5.asp>
- Moran, C., & Park, J.-G. (1986). *Merchandise trade deflators for developing countries* (Division Working Paper No. 1986-7). World Bank, Economic Analysis and Projections Department. <https://documents1.worldbank.org/curated/en/212361468740146509/txt/multi0page.txt>
- The Pennsylvania State University. (2018). *10.7 - Detecting multicollinearity using variance inflation factors* [Course materials]. STAT 462: Applied Regression Analysis. <https://online.stat.psu.edu/stat462/node/180/>
- Singer, D. A., Berger, V. I., & Moring, B. C. (2008). *Porphyry copper deposits of the world: Database and grade and tonnage models, 2008* (U.S. Geological Survey Open-File Report 2008-1155). U.S. Geological Survey. <https://mrdata.usgs.gov/porcu/>
- U.S. Geological Survey. (n.d.-a). *Copper statistics and information*. Retrieved June 13, 2026, from <https://www.usgs.gov/centers/national-minerals-information-center/copper-statistics-and-information>
- U.S. Geological Survey. (n.d.-b). *EarthExplorer*. Retrieved June 13, 2026, from <https://earthexplorer.usgs.gov/>
- U.S. Geological Survey. (n.d.-c). *Mineral Resources Online Spatial Data Access Tool*. Retrieved June 13, 2026, from <https://www.usgs.gov/tools/mineral-resources-online-spatial-data-access-tool>
- U.S. Geological Survey. (n.d.-d). *3D Elevation Program*. Retrieved June 13, 2026, from <https://www.usgs.gov/3d-elevation-program>
- U.S. Geological Survey. (n.d.-e). *USGS EROS Archive—Digital elevation—Shuttle Radar Topography Mission (SRTM) 1 arc-second global*. Retrieved June 13, 2026, from <https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-shuttle-radar-topography-mission-srtm-1>
- World Bank. (1997). *Commodity prices* [Table 6.5, technical notes]. In *World development indicators 1997*. https://www.rrojasdatabank.info/wdi1997/TB6_5.HTM

Final Conclusion and Limitations

Viewed together, the three analytical angles reveal a telling story about US's copper dependence on the outside world - in terms of **supply**, US mine output is declining, down from a 1997 peak of about 1.95 Mt to roughly 1.10 Mt in 2024. While its downward movement can possibly be explained by the decline in economic viability

of mining - or the cheaper costs of imports, we have not been able to analyse more of these variables - such as China's exports to the USA or the volumes of import of not refined copper but copper-embedded products. On the **demand** front, US renewable capacity has grown around fourfold since 1990, led increasingly by the high demand of copper components created by solar and wind infrastructure. Our forecasts point to roughly 4.74 Mt of new copper needed from solar and wind alone by 2030 — before grids, EVs and data centres are counted. On the **price** front, copper's post-2020 steep rise is a real trend, not just driven by inflation: after deflation it has risen about 43% against US consumer prices (CPI) and about 147% against global manufactured goods (MUV). Copper price behaves like a global commodity, and every forecast scenario points upward. Exponential growth analysis has not been applied to price - this can be done too. It would also be useful to compare copper prices to those of other critical metals and look at other background forces driving them - which was not covered in the scope of this project. We were also not able to look at prominent international or USA policies around critical metals or significant supply chain disruption events which could have caused the huge shocks observed in the prices data. We were also limited by a small time range of data availability - since all of our time ranges had to overlap.